

INTEGRAL SECOND-ORDER APÉRY-LIKE RECURRENCES AND RATIONAL ELLIPTIC SURFACES

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ABSTRACT. We study the integral solutions of three-term recurrences $(n+1)^2 u_{n+1} = P(n)u_n - Q(n)u_{n-1}$, $\deg P, \deg Q \leq 2$, in two tiers. *Theorem A* is a conditional classification and is proved: if the Picard–Fuchs operator of such a row is that of an elliptic surface over $\mathbb{P}_t^1$, then for the Zagier shape $(an^2 + an + b, cn^2)$ all four singular fibres are semistable, $\deg \mathcal{J} = 12$ exactly — by Riemann–Hurwitz together with a twist-parity constraint $12 \mid \deg \mathcal{J}$ — and the configuration is, up to quadratic twist, one of Beauville’s six; no hypothesis on the Kodaira dimension is needed, since a twist raises χ (two branch points give a $K3$) without changing $\mathcal{J}$, the local system, or the row. The row is the period expansion at a cusp in the integral coordinate, and the complete list of cusp placements is Zagier’s seven rows **A–G**, with **A, C, F** the three placements of $I_6 I_3 I_2 I_1$ and **E, G** the two of $I_1 I_1 I_2 I_8$. The correspondence is Zagier’s (2009, §7) with Beauville’s theorem; new here are the mechanism — an exponent dictionary, Kodaira admissibility, a cross-ratio invariant, and a cusp-move theorem enumerating the placements — and the completeness of the list. We prove the analogue for the root-row shape $(An^2 + \frac{A}{2}n + B, C(2n-1)^2)$: there $\deg \mathcal{J} = 6$, three of Herfurtnier’s configurations survive up to twist, and exactly two integral rows realise them. Riemann–Hurwitz alone does not force $\chi = 1$ in either shape — in the root-row case a mixed III/III^* elliptic $K3$ is inside the bound — and it is the twist parity, not the degree bound, that settles it. Beukers’ row, the square root of Apéry’s $\zeta(3)$ recurrence, is not among them — it lives on $X_0(6)/W_6$, over which the universal curve does not descend. *Theorem B* collects the unconditional necessary conditions — Kodaira admissibility of the exponent differences, integrality rigidity (rational characteristic roots are integers, so the irrationality score is $\leq -k$ on every rational cusp-move orbit), and the Riemann–Hurwitz bound — which exclude the square roots of Cooper’s s_{10}, s_{18} and an infinite arctan–Padé family. The unconditional classification is exactly Zagier’s conjecture that any integral solution has a modular parametrisation, i.e. Bombieri–Dwork in rank two with four singular points; we make the reduction precise and give the evidence: exhaustive scans (1.27×10^6 integrality hits over 26 Kodaira-admissible classes; 3.0×10^6 eta-quotient pairs), two new elliptic-surface rows on $I_1 I_7 IIII$ and $I_3 IIIIIIII$ (the latter with Chowla–Selberg period $\text{Im } \xi = -\Gamma(1/4)^4 / (2^{7/2} 3^{11/4} \pi)$), and the fact that Apéry’s $\zeta(2)$ row and Beukers’ row are the only rows in the box with positive score and denominator exponent $k \geq 2$.

1. INTRODUCTION

The question. Fix the three-term recurrence

$$(n+1)^2 u_{n+1} = P(n)u_n - Q(n)u_{n-1}, \quad u_0 = 1, \quad u_{-1} = 0, \quad (1)$$

with $P = an^2 + bn + c$ and $Q = dn^2 + en + f$ in $\mathbb{Q}[n]$. Which choices of (P, Q) give $u_n \in \mathbb{Z}$ for all n ?

The question is Apéry’s, in the form Zagier gave it. Apéry’s proof of the irrationality of $\zeta(2)$ runs on $(n+1)^2 u_{n+1} = (11n^2 + 11n + 3)u_n + n^2 u_{n-1}$, whose two solutions are integral, respectively integral after multiplication by $\text{lcm}(1, \dots, n)^2$. Zagier [20] searched the sub-family $P = an^2 + an + b$, $Q = -cn^2$ by computer and found, besides degenerate and hypergeometric solutions, exactly six sporadic integral rows **A, ..., F** (Apéry’s is **D**), all modular: each is the t -expansion of a weight-one form against a Hauptmodul on a genus-zero group, and the six groups are the six semistable configurations of Beauville [2]. Almkvist–Zudilin [1] carried out the analogous search one order up, Cooper [8] added s_7, s_{10}, s_{18} , and Beukers [3] supplied the framework in which all of these are Eichler integrals — and,

in his Theorem 3, an irrationality proof for a period attached to the *square root* of Apéry’s $\zeta(3)$ recurrence.

Zagier’s own §7 already contains the answer to the second half of this, and it should be quoted before anything is claimed. He observes that Beauville’s six stable families give Picard–Fuchs equations of the form (1), that two of Beauville’s families (V and VI) are isogenous to two others (II and I) and give the same equations, that his hypergeometric solutions #2, #11, #14 are also on Beauville’s list, and that “each of Beauville’s surfaces can give rise to several different Picard–Fuchs equations, depending on how one chooses the Hauptmodul $t(z)$. . . we must put one fibre at $t = 0$ and one at ∞ , giving 12 choices”. He then gives the correspondence

Beauville family:	IV	I, VI	III	II, V
Picard–Fuchs equation:	A, C, F	B, #14	D	E, G, #11

and notes $t_{\mathbf{C}}(z) = t_{\mathbf{A}}(z)/(1 + t_{\mathbf{A}}(z))$. So the fact that **A, C, F** are three placements on one surface, and that **E, G** are two on another, is Zagier’s, not ours.

What is not settled there is why the answer is finite, why six, and whether a solution can exist outside the imposed normalisation. Zagier says as much: after listing 12 modular solutions he writes that a complete classification of the *modular* solutions “probably can be done”, that the most optimistic guess is that the 12 already found are complete, and that the passage from “modular” to “all” is the content of

Conjecture (Zagier [20, p. 6]). *Any integral solution of the differential equation (1), where $P(t)$ is a non-degenerate quadratic polynomial, has a modular parametrization.*

What this paper does. We give a two-tier answer. The classification splits into a *local* question — which shapes of (P, Q) are possible — and a *global* one — which shapes actually carry an elliptic surface. Both are settled by proof, conditionally on the geometric hypothesis; the removal of that hypothesis is Zagier’s conjecture, and §5.8 makes the reduction precise.

Theorem A (conditional classification). [PROVED] (given Beauville’s and Herfurtners’ classifications; the explicit lists are exact finite computations, marked as such in §5.1).

- (1) Zagier shape. Let $a, b, c \in \mathbb{Z}$ with $c \neq 0$ and $a^2 \neq 4c$, and let $u_n \in \mathbb{Z}$ solve $(n+1)^2 u_{n+1} = (an^2 + an + b)u_n - cn^2 u_{n-1}$, $u_0 = 1$, $u_{-1} = 0$. Suppose the local system $\ker L$ is, up to a rational gauge factor and a quadratic character twist, $R^1 \pi_* \mathbb{Q}$ of an elliptic surface $\pi : \mathcal{E} \rightarrow \mathbb{P}_t^1$ with non-constant $\mathcal{J}$. Then (a) all four local exponent differences vanish, so all four singular fibres are semistable (up to the twist); (b) $\deg \mathcal{J} \leq 12$ by Riemann–Hurwitz and $12 \mid \deg \mathcal{J}$ by twist parity (Lemma 4.3), so $\deg \mathcal{J} = 12$ exactly and $\mathcal{J}$ is a Belyi map; the untwisted member of the surface’s quadratic-twist class is then semistable with $\sum e = 12$, i.e. a rational elliptic surface, and is one of Beauville’s six. No hypothesis on the Kodaira dimension is required: a quadratic twist raises χ (two branch points give a K3) without changing $\mathcal{J}$, the local system, or the row; (c) t is the Hauptmodul normalised $t = q_h + O(q_h^2)$ at the MUM cusp and u_n is its t -expansion coefficient sequence — the row is the period expansion at a cusp in the integral coordinate; and (d) the cross-ratio $\mathcal{I} = a^2/c$ lies in $\{0, 3, \frac{9}{2}, -\frac{49}{8}, \frac{100}{9}, \frac{289}{72}, -121\}$, the configuration together with the choice of MUM cusp and of the cusp at ∞ determines the operator and hence b , and the complete list of such placements is the seven rows **A–G** of Table 1: three placements of $I_6 I_3 I_2 I_1$ (**A, C, F**), two of $I_1 I_1 I_2 I_8$ (**E, G**), one each of $I_1 I_1 I_1 I_9$ (**B**) and $I_1 I_1 I_5 I_5$ (**D**).
- (2) Root-row shape. Let the row be $(An^2 + \frac{A}{2}n + B, C(2n-1)^2)$ with $A, B, C \in \mathbb{Z}$, integral after a λ^n rescaling, and make the same geometric hypothesis. Then $\rho = \frac{1}{2}$ and $\delta_\infty = 0$: the two finite fibres are of type III or III^* and the fibre at ∞ is a cusp. Riemann–Hurwitz gives $\deg \mathcal{J} \leq 6$ and twist parity gives $\deg \mathcal{J} \equiv 6 \pmod{12}$, so $\deg \mathcal{J} = 6$; the untwisted member of the twist class has fibres I_n, I_m, III, III with $n + m = 6$ and $\sum e = 12$, hence is rational, and is one of Herfurtners’ #38 $I_1 I_5 III III$, #39 $I_2 I_4 III III$, #40 $I_3 I_3 III III$, with $\mathcal{I} = \frac{484}{125}, 0, -12$. Riemann–Hurwitz alone does not force $\chi = 1$ here — the mixed configuration III at t_1 , III^*

at t_2 , I_n at 0, I_m^* at ∞ with $n + m = 6$ has $\sum e = 24$, an elliptic $K3$, and $\deg \mathcal{J} = 6$ sits inside the bound — but that configuration is the quadratic twist with branch set $\{t_2, \infty\}$ of the corresponding #38/#39/#40 one, so it carries the same $\mathcal{J}$, the same local system and the same row (§5.1). Exactly two integral rows realise them: $\sqrt{\text{AZ}(11, 5, 125)} = (88, 10, 500)$ on #38 with cusp width $h = 1$, and $\sqrt{\text{AZ}(9, 3, -27)} = (72, 6, -108)$ on #40 with $h = 3$; the value $\mathcal{I} = 0$ forces $A = 0$ and yields no non-degenerate integral row. In particular Beukers' row $(136, 10, 4)$, with $\mathcal{I} = 1156$, satisfies the hypothesis for no elliptic surface: its projective monodromy is the Atkin–Lehner quotient $\Gamma_0(6) + 6$, and over $\mathbb{P}_t^1 = X_0(6)/W_6$ the universal elliptic curve does not descend — only its symmetric square does.

Part (1) is, in substance, Zagier §7 plus Beauville, and we claim no priority for the correspondence. What is new is the mechanism and the completeness: the exponent dictionary and the fact that Zagier's normalisation is *forced* by semistability rather than assumed (Theorem 2.2); Kodaira admissibility (Theorem 2.3); the cross-ratio invariant $\mathcal{I} = a^2/d$ as a finite *necessary* condition, which turns Zagier's “12 choices” into a finite check (Theorem 3.2); a cusp-move theorem enumerating the placements as a group action, proving the placement list complete and showing that **A**, **C**, **F** carry *two* distinct extension classes on the one surface (§6); the root-row case, which is not in Zagier; the Riemann–Hurwitz bound removing “rational” from the hypothesis (Proposition 4.2); the scans; and the non-congruence classification of §7. Relative to Movasati–Reiter [15], who list the Heun equations attached to Herfurtners' families, we add the recurrence and integrality side, the cross-ratio criterion, the root rows, and the scans.

Theorem B (unconditional necessary conditions). [PROVED] *Let (1) be integral with $d \neq 0$, $a^2 \neq 4d$ and $Q(n) \neq 0$ for $n \geq 1$. Then, with no geometric hypothesis:*

- (i) (normalisation) *if $a^2 - 4d \notin \mathbb{Q}^{\times 2}$ then the two movable exponents agree, $\rho_1 = \rho_2 = \rho$, and $b = (1 - \rho)a$, $e = -2\rho d$;*
- (ii) (Kodaira admissibility) *if $\ker L$ is, up to rational gauge and quadratic twist, that of an elliptic surface, then every local exponent difference lies in $\{0, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}\} \bmod 1$; contrapositively, any row violating this comes from no elliptic surface, in any gauge;*
- (iii) (integrality rigidity) *λ_1, λ_2 are algebraic integers, so $\lambda_1 \in \mathbb{Q}$ iff $\lambda_1, \lambda_2 \in \mathbb{Z}$ iff $a^2 - 4d$ is a square; in that case all three gaps $|\lambda_1|, |\lambda_2|, |\lambda_1 - \lambda_2|$ are positive integers and hence $|\lambda_2| \geq 1$ at every cusp placement, so $\text{score} = \log(1/|\lambda_2|) - k \leq -k \leq -1$ throughout the rational orbit. Consequently a positive-score row has irrational characteristic roots;*
- (iv) (Riemann–Hurwitz) *for any elliptic surface of any Kodaira dimension whose four singular fibres sit over the row's singular points, $\deg \mathcal{J} \leq 6c + 4e_3 + 3e_2 - 12 \leq 12$, where c, e_2, e_3 count the points whose projective local monodromy is unipotent, of order 2, of order 3.*

Corollary C. [PROVED] *The Sym^1 square roots of Cooper's s_{10} and s_{18} ($\delta_\infty = \frac{1}{4}$) and the infinite family $(\alpha, -\frac{\alpha}{6}, -\frac{\alpha}{3}, -27, 63, -30)$, integral exactly for $\alpha \equiv 18 \pmod{36}$ ($\rho = \frac{7}{6} \equiv \frac{1}{6}$), are not Picard–Fuchs systems of any elliptic surface, in any gauge. The second family has unbounded positive score and Apéry limit $-\frac{1}{2\sqrt{3}} \arctan \frac{6\sqrt{3}}{\alpha}$: it is Padé approximation, not geometry.*

The third tier is computational and stands in for the missing converse.

Theorem D (the scans). [VERIFIED: boxes as stated] *An exhaustive integrality scan of all 26 Kodaira-admissible normalisation classes over $1 \leq A \leq 3000$, $|B| \leq 150$, $|C| \leq \min(4000, 12000/M^2)$ — 1 270 065 hits, integrality prime-tested to $n = 30$ and re-verified exactly to $n = 300$ — returns 28 primitive, Casoratian-non-degenerate rows in the rigid classes, of which exactly ten have an elliptic-surface local system (Zagier's six, the two root rows of Theorem A(2), and two new rows on $I_1 I_7 III$ with monodromy $\Gamma_0(7)$ and on $I_3 III III III$); in the family classes the integral rows form infinite families; and the only rows anywhere in the scan with positive score and $k \geq 2$ are Apéry's $\zeta(2)$ row (+0.40606) and Beukers' row (+0.13920). An independent modular scan of 3 036 854 eta-quotient pairs (t, F) on $\Gamma_0(N)$, $N \leq 60$, produces exactly two classes with $|\lambda_2| < 1$: Beukers' and the level-8 T row.*

See Theorem 5.3 and Corollary 5.4 for the details.

Two further structural results are used throughout and stated in full below: the cusp-move theorem (Theorem 6.1), which makes “which fibre sits at ∞ ” into a group action and supplies Theorem B(iii); and the host-group theorem (Theorem 7.3), by which a three-term row forces its host to have exactly four special points, hence covolume index ≤ 12 , so that inside $\mathrm{PSL}_2(\mathbb{Z})$ there are exactly 28 candidate groups, 16 congruence and 12 non-congruence — with Beauville’s six recovered as the four-cusp case. Beukers’ Theorem 3 row is the unique positive-score row whose form is non-congruence.

The honest statement. Theorem A classifies the rows whose operator is *assumed* to be of geometric origin. An unconditional classification — “every integral solution is on the list” — needs the converse, and the converse is exactly Zagier’s conjecture: it is the rank-two, four-singular-point case of the Bombieri–Dwork conjecture that a globally nilpotent operator comes from geometry. One direction of the reduction is proved here: by Theorem 7.3 a modular parametrisation of a three-term row forces four special points, hence one of 28 genus-zero groups, hence (with Theorems A and 7.4) the list. So *Zagier’s conjecture for this class is equivalent to the statement that the projective monodromy group of an integral three-term row is commensurable with $\mathrm{PSL}_2(\mathbb{Z})$* (Proposition 5.6). That statement is not accessible by rigidity — rank two with four singular points is not rigid in Katz’s sense, the accessory parameter being the modulus — nor by the naive implication “globally nilpotent $\Rightarrow$ hypergeometric pullback”, for which Krammer’s Lamé example [13, 15] is the standing warning; the best partial result we know is Beukers’ p -adic analysis of Dwork’s accessory parameter problem in exactly this configuration [4], which does not close it. §5.8 gives the evidence in place of a proof. [OPEN]

Status discipline. Every statement carries its evidence class: [PROVED] (a proof or proof sketch is given), [VERIFIED: R] (an exact finite computation over the stated range R — not a proof, and never silently upgraded), [NUMERICAL: D] (a floating-point or p -adic measurement to D digits), [OPEN]. §9 names the source documents and scripts for every computation.

2. THE EXPONENTS OF A SECOND-ORDER ROW

Let $\theta_t = t \, d/dt$. Substituting $y = \sum_{n \geq 0} u_n t^n$ into (1) gives the Picard–Fuchs operator

$$L = \theta_t^2 - t P(\theta_t) + t^2 Q(\theta_t + 1) = t^2 R(t) \partial_t^2 + t S(t) \partial_t + (-ct + (d+e+f)t^2), \quad (2)$$

with

$$R(t) = 1 - at + dt^2, \quad S(t) = 1 - (a+b)t + (3d+e)t^2.$$

The characteristic roots of the recurrence are $\lambda_i = 1/t_i$, the roots of $\lambda^2 - a\lambda + d$, where t_1, t_2 are the roots of R .

Lemma 2.1 (exponent dictionary). [PROVED] *Assume $d \neq 0$ and $a^2 \neq 4d$, so that L has exactly four singular points $0, t_1, t_2, \infty$. Put $T(t) := S(t) - tR'(t) = 1 - bt + (d+e)t^2$. Then the local exponents are*

$$(0, 0) \text{ at } t = 0; \quad \left(0, \rho_i = -\frac{T(t_i)}{t_i R'(t_i)}\right) \text{ at } t = t_i; \quad (1 - r_1, 1 - r_2) \text{ at } t = \infty,$$

r_1, r_2 being the roots of Q ; and $\rho_1 + \rho_2 = r_1 + r_2 = -e/d$ (Fuchs). The point $t = 0$ is a MUM point: the exponents are equal and the second solution is logarithmic.

Proof. Expanding (2) gives the displayed ∂_t -form. At a simple zero t_i of R the indicial polynomial is $\rho[(\rho-1)t_i R'(t_i) + S(t_i)]$, whence the exponents 0 and $1 - S(t_i)/(t_i R'(t_i)) = -T(t_i)/(t_i R'(t_i))$. At ∞ , $y \sim t^\mu$ forces $Q(\mu+1) = 0$, so the exponents in $s = 1/t$ are $-\mu = 1 - r_i$. The sum of the six exponents is 2, as it must be for a rank-two Fuchsian operator with four singular points on $\mathbb{P}^1$, and equals $\rho_1 + \rho_2 + 2 - (r_1 + r_2)$. $\square$

We write $\delta_\infty := r_2 - r_1$ for the exponent difference at ∞ .

Theorem 2.2 (normalisation classes). [PROVED] *With the hypotheses of Lemma 2.1, $\rho_1 = \rho_2 = \rho$ if and only if*

$$b = (1 - \rho)a \quad \text{and} \quad e = -2\rho d. \quad (3)$$

Writing $Q(n) = C(Mn - j_1)(Mn - j_2)$, so that $r_i = j_i/M$, this reads

$$\rho = \frac{j_1 + j_2}{2M}, \quad \delta_\infty = \frac{j_2 - j_1}{M}, \quad P(n) = A\left(n^2 + \frac{2M - j_1 - j_2}{2M}n\right) + B,$$

with $A = a$, $B = c = u_1$ the accessory parameter and $D = CM^2 = d$.

Proof. $\rho_1 = \rho_2 = \rho$ says $T(t_i) + \rho t_i R'(t_i) = 0$ for $i = 1, 2$, i.e. $R \mid T + \rho t R'$. Both are quadratics with constant term 1, so they are equal; comparing coefficients gives $b + \rho a = a$ and $e + 2\rho d = 0$. $\square$

Modulo the scaling $t \mapsto t/\mu$, $u_n \mapsto \mu^n u_n$ (which sends $(a, b, c, d, e, f) \mapsto (\mu a, \mu b, \mu c, \mu^2 d, \mu^2 e, \mu^2 f)$), a normalisation class is fixed by the pair (ρ, δ_∞) ; inside it the free parameters are A , the leading coefficient C of Q , and the accessory parameter $B = u_1$. A rank-two Fuchsian equation with four singular points has exactly one accessory parameter, and this is it.

Two special cases recover the two normalisations that occur in the literature.

- $\rho = 0$ and $\delta_\infty = 0$ give $b = a$, $e = f = 0$: *Zagier's shape* ($an^2 + an + c$, dn^2). Geometrically $\rho = 0$ says both movable fibres are semistable and $\delta_\infty = 0$ says the fibre at ∞ is too. So Zagier's normalisation is not a choice: it is the statement that all four fibres are semistable.
- $\rho = \frac{1}{2}$ and $\delta_\infty = 0$ give $b = \frac{a}{2}$ and $Q = C(2n - 1)^2$: the *root-row shape*, observed empirically for all nine Sym^1 square roots of the third-order sporadic families. Geometrically the two movable fibres have order-two monodromy (*III* or *III**) and the fibre at ∞ has half-integral exponents (I_m^*).

2.1. Kodaira admissibility.

Theorem 2.3. [PROVED] *Suppose the local system $\ker L$ is, up to a rational gauge factor and a quadratic character twist, $R^1\pi_*\mathbb{Q}$ of an elliptic surface $\pi : \mathcal{E} \rightarrow \mathbb{P}_t^1$. Then*

$$\rho_1, \rho_2, \delta_\infty \in \{0, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}\} \pmod{1}.$$

Proof. A rational gauge factor shifts exponents by integers, a quadratic twist by half-integers, at each singular point; neither changes an exponent *difference* mod 1. The exponent difference at a point is $\frac{1}{2\pi i} \log$ of the ratio of the eigenvalues of the local monodromy, and Kodaira's list [12] gives the eigenvalue pairs $(1, 1)$ for I_n , $(-1, -1)$ for I_n^* , $(\pm i)$ for III, III^* , $(e^{\pm 2\pi i/3})$ for IV, IV^* and $(e^{\pm \pi i/3})$ for II, II^* ; the ratios are 1, 1, -1 , $e^{\mp 4\pi i/3}$, $e^{\mp 2\pi i/3}$, i.e. differences 0, 0, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{2}{3}$ mod 1. $\square$

The three admissible values are exactly the three kinds of point of a Fuchsian group commensurable with $\text{PSL}_2(\mathbb{Z})$: a cusp, an elliptic point of order two, an elliptic point of order three. The MUM point is always a cusp.

Corollary 2.4. [PROVED] *The following are not Picard–Fuchs systems of an elliptic surface, in any gauge:*

- the Sym^1 square roots of Cooper's s_{10} and s_{18} , whose trailing coefficients are $-4(8n - 3)(8n - 5)$ and $12(8n - 3)(8n - 5)$, so that $\delta_\infty = \frac{1}{4}$ — an order-eight local monodromy;*
- the one-parameter family $(\alpha, -\frac{\alpha}{6}, -\frac{\alpha}{3}, -27, 63, -30)$, integral exactly for $\alpha \equiv 18 \pmod{36}$, which has $\rho = \frac{7}{6} \equiv \frac{1}{6}$.*

Item (b) is the sharpest available warning against reading a positive score as evidence of a new object: the family is $(n + 1)^2 u_{n+1} = (3n - 2) \left[\frac{\alpha}{6} (2n + 1) u_n + 3(3n - 5) u_{n-1} \right]$ with $\lambda_1 \lambda_2 = -27$ and score $\approx \log(\alpha/27) - 2 \rightarrow \infty$, its member $\alpha = 234$ scoring $+0.15998$, above Beukers' row; and its Apéry limit is $-\frac{1}{2\sqrt{3}} \arctan \frac{6\sqrt{3}}{\alpha}$ [NUMERICAL: 193 digits, $\alpha = 18, 54, 90, 126, 234, 306, 558, 846$], the arctan analogue of the classical Legendre–Padé log family. Padé approximation, not geometry — and Theorem 2.3 says so independently of the closed form.

Corollary 2.5 (Galois rigidity). [PROVED] *If $a^2 - 4d$ is not a square then t_1, t_2 are conjugate irrationals, hence $\rho_1 = \rho_2$ and the two movable fibres have the same Kodaira class. Equivalently: a configuration whose two movable fibres differ in class requires $a^2 - 4d \in \mathbb{Q}^{\times 2}$, so $\lambda_1, \lambda_2 \in \mathbb{Z}$, so $|\lambda_2| \geq 1$ and the score $\log(1/|\lambda_2|) - k$ is $\leq -k < 0$.*

Proof. Let σ be the nontrivial automorphism of $\mathbb{Q}(t_1)$. The conditions $T(t_i) + \rho_i t_i R'(t_i) = 0$ have rational coefficients, so applying σ to the condition at t_1 gives $T(t_2) + \rho_1 t_2 R'(t_2) = 0$; since $t_2 R'(t_2) \neq 0$ this forces $\rho_2 = \rho_1$. For the second clause, $\lambda_{1,2}$ are the roots of the monic integer quadratic $\lambda^2 - a\lambda + d$, hence rational integers when the discriminant is a square; the only integer of absolute value < 1 is 0, and $\lambda_2 = 0$ means $d = 0$, excluded. $\square$

3. HERFURTNER'S LIST AND THE CROSS-RATIO INVARIANT

Herfurtner [10] classifies the minimal elliptic surfaces over $\mathbb{P}^1$ with section, non-constant $\mathcal{J}$, and exactly four singular fibres. Since $e(I_0^*) = 6$, two fibres in $T^- = \{I_n^*, IV^*, III^*, II^*\}$ force $\sum e \geq 14 > 12$, so a *rational* such surface has at most one fibre in T^- ; his Table 3 lists 50 twist classes, and “transfer of $*$ ” ($I_n \leftrightarrow I_n^*$, $II \leftrightarrow IV^*$, $III \leftrightarrow III^*$, $IV \leftrightarrow II^*$) gives the full list of 56 quadruples: 38 *rigid* (all fibres in T^+ ; the surface unique, $\mathcal{J}$ one of 38 Belyi maps — the same list as Vidunas–Filipuk’s Table 4 [19] and Movasati–Reiter’s Table 1 [15]) and 18 in *one-parameter* families. One has $\deg \mathcal{J} = \sum n_i$ over the I_n and I_n^* fibres, and the six semistable configurations $I_9 I_1 I_1 I_1$, $I_8 I_2 I_1 I_1$, $I_6 I_3 I_2 I_1$, $I_5 I_5 I_1 I_1$, $I_4 I_4 I_2 I_2$, $I_3 I_3 I_3 I_3$ are Beauville’s [2], all rigid with $\deg \mathcal{J} = 12$.

A row imposes three conditions on a configuration: a cusp at $t = 0$ (the MUM point); the two movable points t_1, t_2 carry fibres of the *same* Kodaira class unless $a^2 - 4d$ is a square (Corollary 2.5); and the fourth fibre sits at $t = \infty$. Grouping the 38 rigid configurations by the pair (Kodaira class at t_1, t_2 ; class at ∞) gives nine normalisation classes. Beauville’s six occupy (cusp, cusp), i.e. $(\rho; \delta_\infty) = (0; 0)$; (cusp, ord 2) holds #23–#26 ($I_1 I_1 I_7 III$, $I_1 I_2 I_6 III$, $I_1 I_3 I_5 III$, $I_2 I_3 I_4 III$); (cusp, ord 3) holds #19–#22 and #27–#29; (ord 2, cusp) — the root-row class — holds #38 $I_1 I_5 III III$, #39 $I_2 I_4 III III$, #40 $I_3 I_3 III III$; (ord 2, ord 2) holds only #45 $I_3 III III III$; (ord 2, ord 3) holds #44, #48; (ord 3, cusp) holds #30 $I_1 I_7 II II$, #31, #32, #36, #37, #43; (ord 3, ord 2) holds #46, #49; and (ord 3, ord 3) holds #47, #50. [PROVED] (given Herfurtner’s table).

The one-parameter families enter as follows. An I_0^* fibre has trivial image in PSL_2 , so at such a point L has an *apparent* singularity (exponents $(0, m)$ with $m \in \mathbb{Z}_{>0}$ and no logarithm): these are the classes with $\rho \in \mathbb{Z}_{>0}$. An I_m^* has half-integral exponents, hence in our normalisation can only sit at ∞ , and that is precisely what $Q = C(2n - 1)^2$ does. So the root-row class $(\frac{1}{2}; 0)$ meets both the rigid block and, a priori, families with an I_m^* at ∞ ; but $\sum e = I_a + 3 + 3 + (m + 6) = 12$ forces $a + m = 0$, impossible, so in the root-row class only #38, #39, #40 survive.

3.1. The cross-ratio invariant.

Definition 3.1. The row (1) has four singular points $0, t_1, t_2, \infty$, of which 0 is the MUM cusp and $\{t_1, t_2\}$ is an unordered pair. Its *cross-ratio invariant* is

$$\mathcal{I} := \frac{(1+z)^2}{z} = \frac{(\lambda_1 + \lambda_2)^2}{\lambda_1 \lambda_2} = \frac{a^2}{d}, \quad z = \frac{t_1}{t_2} = \frac{\lambda_2}{\lambda_1},$$

invariant under $z \mapsto 1/z$, hence under both allowed relabellings ($t_1 \leftrightarrow t_2$, and $0 \leftrightarrow \infty$ when both are cusps).

For a Herfurtner configuration with base points $(p_1, \dots, p_4)$ and an assignment $(p_0, p_\infty \mid p', p'')$ the same quantity is $\mathcal{I} = (1+z)^2/z$ with $z = \frac{(p' - p_0)(p'' - p_\infty)}{(p' - p_\infty)(p'' - p_0)}$.

Theorem 3.2 (finiteness). [PROVED] (given Herfurtner’s table) *A row (1) with $a, d \in \mathbb{Z}$ is the Picard–Fuchs system of a rigid Herfurtner configuration only if $\mathcal{I} = a^2/d$ equals the cross-ratio invariant of that configuration under an assignment compatible with Lemma 2.1. Since $\mathcal{I} \in \mathbb{Q}$, every configuration*

whose invariant is irrational or non-real is excluded outright. Per class the admissible sets are finite and explicit; in particular

$$\text{all four fibres semistable: } \mathcal{I} \in \left\{0, 3, \frac{9}{2}, -\frac{49}{8}, \frac{100}{9}, \frac{289}{72}, -121\right\},$$

$$\text{root-row class } (\tfrac{1}{2}, 0): \mathcal{I} \in \left\{\frac{484}{125} (\#38), 0 (\#39), -12 (\#40)\right\}, \quad (\tfrac{1}{2}; \tfrac{1}{2}): \mathcal{I} = 3 (\#45),$$

$$(\tfrac{1}{3}; 0), (\tfrac{2}{3}; 0): \mathcal{I} \in \left\{\frac{169}{49}, 0, -50, \frac{1681}{400}, -\frac{49}{8}\right\}.$$

Proof. $\mathcal{I}$ is a Möbius invariant of the labelled quadruple, and the identification of $\ker L$ with the surface's local system is by a Möbius map of $\mathbb{P}_t^1$ matching the four singular points to the four base points, in a way compatible with the Kodaira classes read off from Lemma 2.1. For each of the finitely many rigid configurations and each of the finitely many compatible assignments one computes $\mathcal{I}$ from the base points; the displayed lists are the outcome. $\square$

The eighth semistable value is $65 + \frac{682}{25}\sqrt{5}$, belonging to $I_5I_5I_1I_1$ with the two I_1 's taken as the movable pair; it is irrational, hence unreachable by an integer recurrence.

3.2. Zagier's six, read off the table. The six rows **A**(7, 2, -8), **B**(9, 3, 27), **C**(10, 3, 9), **D**(11, 3, -1), **E**(12, 4, 32), **F**(17, 6, 72) have $\mathcal{I} = a^2/c = -\frac{49}{8}, 3, \frac{100}{9}, -121, \frac{9}{2}, \frac{289}{72}$, matching #15 $I_1I_2I_3I_6$ (three times), #13 $I_1I_1I_1I_9$, #16 $I_1I_1I_5I_5$ and #14 $I_1I_1I_2I_8$, all with MUM fibre I_1 and $\deg \mathcal{J} = 12$ (Table 1). The matching is Zagier's [20, §7], obtained there by computing the j -invariant of each Beauville family as a rational function of t and inverting (the calculations are in Verrill [17]); what the cross-ratio adds is that the match can be made, and its completeness checked, without computing j at all. The invariant alone does not separate #13 from #18 (both $\mathcal{I} = 3$) nor #14 from #17 (both $\frac{9}{2}$); the $\mathcal{J}$ -map test of §4 does, by returning the width h of the MUM cusp: $h = 1$ for **B** (so the MUM fibre is I_1 , hence #13, since #18 = $I_3I_3I_3I_3$ has all cusps of width 3) and $h = 1$ for **E** (#14, not #17 whose widths are 2, 2, 4, 4). [VERIFIED: cross-ratios exact; $\mathcal{J}$ -fits exact]

The most interesting line is that #15 occurs three times: **A**, **C**, **F** are three rows on the *same* surface with the *same* MUM fibre, differing only in which of I_2, I_3, I_6 is placed at ∞ . This is Zagier's table entry " $IV \mapsto \mathbf{A}, \mathbf{C}, \mathbf{F}$ " together with his remark $t_{\mathbf{C}} = t_{\mathbf{A}}/(1 + t_{\mathbf{A}})$; §6 makes it an identity of operators with an explicit gauge, adds the second placement pair $\{\mathbf{E}, \mathbf{G}\}$, proves the placement list complete, and — this part is not in [20] — shows that the three rows on the one surface carry *two* different extension classes.

3.3. Why "six". Read the classification backwards. Fix all four fibres semistable. Then Theorem 2.2 forces the shape $(an^2 + an + b, cn^2)$; Herfurtnner (here = Beauville) gives six surfaces, of which V and VI are isogenous to II and I and give the same equations [20, §7]; Theorem 3.2 converts each configuration and placement into a value of a^2/c ; seven of the eight available values are rational, and each rational value is realised by exactly one integral row. The eighth, $65 + \frac{682}{25}\sqrt{5}$, is irrational and therefore unreachable by an integer recurrence. The value $\mathcal{I} = 0$ needs $a = 0$ and gives Zagier's **G**: $u_{2m} = \binom{2m}{m}^2$, $u_{2m+1} = 0$, whose characteristic roots ± 4 have equal modulus so that there is no Apéry limit — which is exactly why Zagier's *sporadic* list has six entries and the placement list has seven. Hence:

The "six" of Beauville–Zagier is, on the recurrence side, the number of rational cross-ratios in the four-cusp block of Herfurtnner's table that carry a non-degenerate integral accessory parameter.

The two Beauville configurations that do not appear as I – VI labels in Table 1 are $I_4I_4I_2I_2$ and $I_3I_3I_3I_3$; they are the isogenous partners V and I of II and VI , their only rational invariants are $\mathcal{I} = 0$ and $\mathcal{I} = 3$, and those values are already taken by **G** and **B**.

4. THE FUNCTIONAL INVARIANT AND THE $K3$ QUESTION

The cross-ratio test is necessary but not sufficient: it uses the positions of the singular points and ignores the accessory parameter. A sufficient test comes from the nome.

Lemma 4.1 (the $\mathcal{J}$ -map test). [PROVED] *Let $q = t \exp(g/y_0)$ be the canonical nome of L at the MUM point, where $y_1 = y_0 \log t + g$ is the second Frobenius solution. If t is a Hauptmodul of a genus-zero group Γ with $t = c q_h + O(q_h^2)$ at a cusp of width h , $q_h = e^{2\pi i \tau/h}$, then $q = c q_h$, hence $e^{2\pi i \tau} = (q/c)^h$. Therefore Γ is conjugate into $\mathrm{PSL}_2(\mathbb{Z})$ — equivalently L is projectively the Picard–Fuchs system of an elliptic surface over $\mathbb{P}_t^1$ — if and only if for some $h \in \mathbb{Z}_{>0}$ and some constant $\gamma = c^{-h}$ the composite $J(\gamma q(t)^h)$, with $J = q^{-1} + 744 + 196884q + \dots$ the classical modular function, is a rational function of t ; and then $\deg \mathcal{J} = \deg J(t)$.*

In practice one forms $W := t^h J(\gamma q(t)^h) \in \mathbb{Q}[[t]]$ and solves $V \cdot W = t^h U$ with $\deg U, \deg V \leq d_{\max}$ by linear algebra over $\mathbb{Q}$, checking the fit on more coefficients than were used to produce it. A *positive* verdict is a certificate; a *negative* verdict is only “no fit within the stated window”, which is why the next proposition matters.

Proposition 4.2 (the window is never the constraint). [PROVED] *Let $\pi : \mathcal{E} \rightarrow \mathbb{P}_t^1$ be a minimal elliptic surface with section and non-constant functional invariant $\mathcal{J}$, $\mu := \deg \mathcal{J}$, whose singular fibres sit over the four singular points $0, t_1, t_2, \infty$ of the row’s operator. Among those four points let c be the number carrying an I_n or I_n^* fibre ($n \geq 1$; a cusp), e_2 the number carrying III or III^* , e_3 the number carrying II, IV, IV^*, II^* . Then*

$$\deg \mathcal{J} \leq 6c + 4e_3 + 3e_2 - 12 \leq 12,$$

with equality on the right only in the all-cusp class. This holds for a surface of any Kodaira dimension — rational, $K3$, or higher.

Proof. By Kodaira’s dictionary for the functional invariant (see Miranda [14], ch. IV): $\mathcal{J}^{-1}(\infty)$ is exactly the set of I_n/I_n^* fibres, with ramification index n , so $\#\mathcal{J}^{-1}(\infty) = c$ and $\sum_{\mathcal{J}(P)=\infty} (e_P - 1) = \mu - c$; over $j = 0$ the ramification index is divisible by 3 except at fibres of type II, IV, IV^*, II^* , so at most e_3 preimages have $e_P < 3$ and $\sum_{\mathcal{J}(P)=0} (e_P - 1) \geq \frac{2}{3}(\mu - e_3)$; over $j = 1728$ the index is even except at III, III^* , giving $\sum_{\mathcal{J}(P)=1728} (e_P - 1) \geq \frac{1}{2}(\mu - e_2)$. Riemann–Hurwitz for $\mathcal{J} : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ gives

$$2\mu - 2 = \sum_P (e_P - 1) \geq (\mu - c) + \frac{2}{3}(\mu - e_3) + \frac{1}{2}(\mu - e_2) = \frac{13}{6}\mu - c - \frac{2}{3}e_3 - \frac{1}{2}e_2,$$

i.e. $\frac{1}{6}\mu \leq c + \frac{2}{3}e_3 + \frac{1}{2}e_2 - 2$, which is the bound. Any further ramification only strengthens it. Maximising $6c + 4e_3 + 3e_2 - 12$ subject to $c + e_2 + e_3 \leq 4$ gives 12, attained only at $c = 4$. Fibres of type I_0^* have trivial image in PSL_2 , are invisible to L , and change nothing. $\square$

This is the genus formula $g = 1 + \frac{\mu}{12} - \frac{e_2}{4} - \frac{e_3}{3} - \frac{c}{2}$ for a genus-zero Fuchsian group of index μ and signature $(0; 2^{e_2}, 3^{e_3}; c)$, written so that it survives extra ramification, i.e. so that it does not assume t is a Hauptmodul. Class by class the bound reads ≤ 12 for $(0; 0)$, ≤ 6 for $(\frac{1}{2}; 0)$, ≤ 8 for $(\frac{1}{3}; 0)$, ≤ 4 for $(\frac{1}{2}; \frac{1}{3})$, ≤ 3 for $(\frac{1}{2}; \frac{1}{2})$; and for the three classes whose movable points carry an apparent singularity ($\rho \in \mathbb{Z}_{>0}$) the bound is ≤ 0 , i.e. those classes contain no elliptic-surface row at all, independently of any search.

Riemann–Hurwitz is not the whole story: it bounds $\deg \mathcal{J}$ but says nothing about χ , and a quadratic twist changes χ without changing $\mathcal{J}$. The missing constraint is a parity.

Lemma 4.3 (twist parity). [PROVED] *In the situation of Proposition 4.2, let $e_{\min}(P)$ be the Euler number of the unstarred member of the twist pair at P : n at an I_n/I_n^* , 3 at a III/III^* , 2 at a II/IV^* , 4 at a IV/II^* . Then*

$$12 \mid \sum_P e_{\min}(P) = \deg \mathcal{J} + 3e_2 + \sum_{\text{ord-3 } P} e_{\min}(P),$$

the sum running over the four singular points of the row.

Proof. For a relatively minimal elliptic surface with section over $\mathbb{P}^1$, $\sum e = 12\chi$. Every singular fibre of $\mathcal{E}$ lies either over one of the four points or is an I_0^* (trivial image in PSL_2 , invisible to L ; any I_n^* with $n \geq 1$ elsewhere would be a fifth singular point of L). Fix the reference twist in which every fibre is unstarred; the actual surface is its quadratic twist by a character ε_S with branch set S , and at each point of S the fibre flips to its starred partner, raising e by exactly 6 — for $I_n \rightarrow I_n^*$, $III \rightarrow III^*$, $II \rightarrow IV^*$, $IV \rightarrow II^*$ and $I_0 \rightarrow I_0^*$ alike. Hence $\sum e = \sum_P e_{\min}(P) + 6|S|$. A quadratic character on $\mathbb{P}^1$ has an even number of branch points, so $\sum e \equiv \sum_P e_{\min}(P) \pmod{12}$, and $12 \mid \sum e$. Finally $\sum_P e_{\min}(P) = \deg \mathcal{J} + 3e_2 + \sum_{\mathrm{ord}=3} e_{\min}$, since the cusps contribute their n_i , whose sum is $\deg \mathcal{J}$. $\square$

Proposition 4.2 and Lemma 4.3 together pin $\deg \mathcal{J}$ exactly in every class, and pin the surface up to quadratic twist. Class by class:

class $(\rho; \delta_\infty)$	(c, e_2, e_3)	RH bound	parity	forced $\deg \mathcal{J}$	untwisted configuration
$(0; 0)$	$(4, 0, 0)$	≤ 12	$12 \mid D$	12	Beauville's six
$(\frac{1}{2}; 0)$	$(2, 2, 0)$	≤ 6	$12 \mid D+6$	6	#38, #39, #40
$(\frac{1}{3}; 0), (\frac{2}{3}; 0)$	$(2, 0, 2)$	≤ 8	$12 \mid D+s, s \in \{4, 6, 8\}$	8, 6, 4	#30–#32; #36, #37; #43
$(\frac{1}{2}; \frac{1}{2})$	$(1, 3, 0)$	≤ 3	$12 \mid D+9$	3	#45
$(\frac{1}{2}; \frac{1}{3}), (\frac{1}{2}; \frac{2}{3})$	$(1, 2, 1)$	≤ 4	$12 \mid D+6+e_{\min}$	4 or 2	#48; #44

In every case the untwisted configuration has $\sum e = 12$, i.e. $\chi = 1$: it is a *rational* elliptic surface. In the third row $s = e_{\min}(t_1) + e_{\min}(t_2)$ is 4, 6 or 8 according as t_1, t_2 carry II, II ; II, IV ; or IV, IV , and by Corollary 2.5 the mixed case $s = 6$ needs $a^2 - 4d$ a square. The point of the lemma is that the parity, not Riemann–Hurwitz, is what excludes the smaller degrees: in the all-cusp class RH alone permits $\deg \mathcal{J} = 6$ (with one extra simple ramification point, i.e. a one-parameter family carrying a single I^* fibre), and it is $12 \mid \deg \mathcal{J}$ that rules it out.

Remark 4.4 (the $K3$ recheck). An elliptic $K3$ with four singular fibres has $\sum e = 24$, so one might expect $\deg \mathcal{J}$ up to 24. Proposition 4.2 says otherwise: the difference between $\sum e = 12$ and 24 is absorbed by fibres contributing to e but not to $\deg \mathcal{J}$ ($II, III, IV, IV^*, III^*, II^*$ contribute 2, 3, 4, 8, 9, 10 to $\sum e$ and 0 to $\deg \mathcal{J}$), not by a larger $\mathcal{J}$; equivalently, by Lemma 4.3 a $K3$ realisation is a quadratic twist of a rational one and shares its $\mathcal{J}$. The computation agrees: re-run with $\deg \mathcal{J} \leq 24$, $h \leq 24$, 140 series terms and 992 values of γ per h (against 12, 12, 56, 96) on all 18 negative rows, on three Kodaira-inadmissible controls and on the six known positives, *every negative stays negative and every positive returns the same* ($h, \deg \mathcal{J}$), with 50 coefficients fitted and 65 further identities verified. [VERIFIED: $\deg \leq 24$, $h \leq 24$, 140 terms, 23 808 (h, γ) pairs per row, 27 rows]

Theorem 4.5 (which root rows are elliptic surfaces). [VERIFIED: exact, series order 56 resp. 140, degrees ≤ 12 resp. ≤ 24] *Of the nine Sym^1 square roots of the third-order sporadic families, exactly two are Picard–Fuchs systems of an elliptic surface over their own $\mathbb{P}_t^1$:*

$$\sqrt{\mathrm{AZ}(11, 5, 125)} \leftrightarrow I_1 I_5 III III \text{ (\#38, } \deg \mathcal{J} = 6, h = 1),$$

$$\sqrt{\mathrm{AZ}(9, 3, -27)} \leftrightarrow I_3 I_3 III III \text{ (\#40, } \deg \mathcal{J} = 6, h = 3),$$

in agreement with $\mathcal{I} = \frac{484}{125}$ and $\mathcal{I} = -12$. The remaining seven — Apéry's own square root, i.e. Beukers' Theorem 3 row ($\mathcal{I} = 1156$), the T row (36), Domb's ($\frac{25}{4}$), $\mathrm{AZ}(7, 3, 81)$'s ($\frac{196}{81}$) and Cooper's s_7 's ($-\frac{676}{27}$), together with $\sqrt{s_{10}}, \sqrt{s_{18}}$ which are excluded already by Corollary 2.4 — fail the test: their projective monodromy is an Atkin–Lehner quotient $\Gamma_0(N) + W$, commensurable with but not conjugate into $\mathrm{PSL}_2(\mathbb{Z})$.

The mechanism is transparent on Apéry's curve, where $t = (\eta_1 \eta_6 / \eta_2 \eta_3)^{12}$ has degree two on $X_0(6)$ and is W_6 -invariant, so $\mathbb{P}_t^1 = X_0(6)/W_6$: the universal elliptic curve does not descend along that quotient, only its symmetric square does, and j is not W_6 -invariant, so no $\mathcal{J}$ -map of finite degree exists. That is why Apéry's *order-three* row lives happily on $\mathbb{P}_t^1$ while its square root does not. The two fixed points of W_6 become the two order-two points of the quotient, i.e. the two exponents $\rho = \frac{1}{2}$

row	Zagier #	(a, b, c)	configuration	fibre at $t = \infty$	$\mathcal{I} = a^2/c$
A	#5	$(7, 2, -8)$	$I_6 I_3 I_2 I_1$ (IV)	I_2	$-49/8$
C	#8	$(10, 3, 9)$	$I_6 I_3 I_2 I_1$ (IV)	I_3	$100/9$
F	#13	$(17, 6, 72)$	$I_6 I_3 I_2 I_1$ (IV)	I_6	$289/72$
B	#7	$(9, 3, 27)$	$I_1 I_1 I_1 I_9$ (I, VI)	I_1	3
D	#9	$(11, 3, -1)$	$I_1 I_1 I_5 I_5$ (III)	I_5	-121
E	#10	$(12, 4, 32)$	$I_1 I_1 I_2 I_8$ (II, V)	I_2	$9/2$
G	#2	$(0, 0, -16)$	$I_1 I_1 I_2 I_8$ (II, V)	I_8	0

TABLE 1. The complete list of cusp placements in the Zagier shape (Theorem A(1)(d)), in Zagier’s own labelling **A–G** and with his index numbers; the Roman numerals are Beauville’s family labels as in [20, §7]. The cusp-move orbits are $\{\mathbf{A}, \mathbf{C}, \mathbf{F}\}$, $\{\mathbf{B}\}$, $\{\mathbf{D}\}$, $\{\mathbf{E}, \mathbf{G}\}$ (§6). **G** is $u_{2m} = \binom{2m}{m}^2$, $u_{2m+1} = 0$: integral but with $|\lambda_1| = |\lambda_2| = 4$, hence no Apéry limit, which is why Zagier’s *sporadic* list stops at six. Zagier’s remaining modular solutions #11 $(16, 4, 0)$, #14 $(27, 6, 0)$ and #20 $(64, 12, 0)$ have $c = 0$ and only three singular points, so they fall outside the hypothesis and belong to Schmickler-Hirzebruch’s three-fibre classification [16].

that the root-row shape records; one is the Fricke fold point $i/\sqrt{6}$, with $t(i/\sqrt{6}) = (\sqrt{2} - 1)^4$ exactly. Same story for T (W_8 on $X_0(8)$), Domb and AZ(7, 3, 81).

Remark 4.6. The $\mathcal{J}$ -test is not implied by the cross-ratio test. Row #14 of Table 2 has $\mathcal{I} = -50$, the invariant of Herfurtnier #32 $I_4 I_4 III$, yet the $\mathcal{J}$ -test says no: the cross-ratio ignores the accessory parameter. Making the test sufficient would require computing the accessory parameter of each Herfurtnier Weierstrass model from its G_2, G_3 and comparing — mechanical, and [OPEN] here.

5. THE TWO-TIER CLASSIFICATION, AND THE SCAN

5.1. Proof of Theorem A. *Part (1).* The Zagier shape has $b = a$ and $e = f = 0$, so by Lemma 2.1 $\rho_1 = \rho_2 = 0$ and $r_1 = r_2 = 0$, i.e. all four exponent differences vanish. By Theorem 2.3 the projective local monodromy at each of the four points is unipotent, so each singular fibre is of type I_n or I_n^* ; this is (a). For (b), Proposition 4.2 with $(c, e_2, e_3) = (4, 0, 0)$ gives $\deg \mathcal{J} \leq 12$, and Lemma 4.3 with $e_2 = e_3 = 0$ gives $12 \mid \deg \mathcal{J}$; each of the four points carries a nontrivial unipotent projective monodromy, so $n_i \geq 1$ and $\deg \mathcal{J} = \sum n_i \geq 4$. Hence $\deg \mathcal{J} = 12$. Equality in Riemann–Hurwitz then forces $\mathcal{J}$ to be unramified outside $\{0, 1728, \infty\}$ with the minimal indices there — a Belyi map — and the untwisted member of the twist class has $\sum e = \sum n_i = 12$, i.e. $\chi = 1$: a rational semistable elliptic surface with four singular fibres, one of Beauville’s six [2]. The surface itself is that one up to quadratic twist, so $\chi = 1 + |S|/2$ may be 2 (a $K3$) or more; but the twist changes neither $\mathcal{J}$, nor the local system, nor the row, which is why no hypothesis on the Kodaira dimension is needed and why the $K3$ range produces nothing new. Riemann–Hurwitz alone would leave $\deg \mathcal{J} = 6$ open, with one extra simple ramification point and a single I^* fibre — exactly Herfurtnier’s one-parameter families; it is the parity $12 \mid \deg \mathcal{J}$ that excludes it. For (c), the MUM point is a cusp and t pulls back the canonical nome (Lemma 4.1); normalising $t = q_h + O(q_h^2)$ makes t a Hauptmodul with integral q -expansion, and $u_n = [t^n]F$ for the weight-one form F attached to the fibration. For (d), $\mathcal{I} = a^2/c$ is a Möbius invariant of the labelled quadruple, so it must be the invariant of the configuration under the chosen placement; Theorem 3.2 lists the seven rational values available in the four-cusp block. Given a configuration and a placement — which cusp at $t = 0$, which at $t = \infty$ — the rigid surface, and hence its Picard–Fuchs operator, is determined, so the accessory parameter b is determined; the enumeration of the resulting placements is Table 1. [VERIFIED: cross-ratios exact; the $\mathcal{J}$ -fits of §4 exact, 56-term series, degrees ≤ 12 , re-checked at $\deg \leq 24$] $\square$

Part (2). The root-row shape has $b = \frac{a}{2}$ and $Q = C(2n - 1)^2$, so $\rho_1 = \rho_2 = \frac{1}{2}$ and $r_1 = r_2 = \frac{1}{2}$: the movable points have exponent difference $\frac{1}{2}$ and ∞ has exponents $(\frac{1}{2}, \frac{1}{2})$. By Theorem 2.3 the

movable fibres are III or III^* and the fibre at ∞ is a cusp I_m or I_m^* , with $m \geq 1$: $m = 0$ would make ∞ an I_0^* , invisible to L , and Proposition 4.2 with $(c, e_2, e_3) = (1, 2, 0)$ would then give $\deg \mathcal{J} \leq 0$. Likewise $n \geq 1$ at the MUM point. Now Proposition 4.2 with $(c, e_2, e_3) = (2, 2, 0)$ gives $\deg \mathcal{J} = n + m \leq 6 \cdot 2 + 3 \cdot 2 - 12 = 6$, while Lemma 4.3 gives $12 \mid n + m + 6$, i.e. $\deg \mathcal{J} \equiv 6 \pmod{12}$. With $2 \leq \deg \mathcal{J} \leq 6$ this forces $\deg \mathcal{J} = 6$, and the untwisted member of the twist class has fibres I_n, I_m, III, III with $n + m = 6$ and $\sum e = 6 + 3 + 3 = 12$: a rational elliptic surface, the three possibilities $(n, m) = (1, 5), (2, 4), (3, 3)$ being Herfurtner's #38, #39, #40 with cross-ratios $\frac{484}{125}, 0, -12$.

It is worth being explicit about what Riemann–Hurwitz does *not* do here. Euler numbers alone leave $\chi = 2$ open. Writing σ for the number of stars among $\{0, \infty\}$: with III at both movable points $\sum e = n + m + 6 + 6\sigma = 12\chi$, and $\deg \mathcal{J} = n + m \leq 6$ still admits $(\sigma, \chi) = (2, 2)$; with III^* at both, $n + m = 12\chi - 24 - 6\sigma$ leaves nothing new; and in the *mixed* case III at t_1 , III^* at t_2 — indistinguishable at the level of exponents, both having difference $\frac{1}{2}$ — one gets $n + m = 12\chi - 18 - 6\sigma$, so $(\sigma, \chi) = (1, 2)$ with $n + m = 6$ is permitted: an elliptic $K3$ with fibres III, III^*, I_n, I_m^* , $n + m = 6$, $\sum e = 3 + 9 + n + (m + 6) = 24$ and $\deg \mathcal{J} = 6$, comfortably inside the bound. *Riemann–Hurwitz does not exclude it*. What removes it as a source of new rows is that it is the quadratic twist with branch set $\{t_2, \infty\}$ of the corresponding #38/#39/#40 configuration — twisting flips $III^* \rightarrow III$ at t_2 and $I_m^* \rightarrow I_m$ at ∞ , lowering $\sum e$ by 12 to $\chi = 1$. The two surfaces have the same $\mathcal{J}$, the same projective monodromy, and local systems differing by the quadratic twist already allowed in the hypothesis; so they carry the same row. Lemma 4.3 is the systematic form of this observation. It is also confirmed computationally: the $\mathcal{J}$ -map test run at $\deg \mathcal{J} \leq 24$, $h \leq 24$ finds no fit for any root row other than the two on #38 and #40 [VERIFIED: 27 rows, 23 808 (h, γ) pairs each], so no $K3$ realisation occurs among the known root rows either.

The realisations, and the negative verdict for the other seven Sym^1 square roots, are Theorem 4.5; the value $\mathcal{I} = 0$ needs $A = 0$, and the scan below finds no non-degenerate integral row there. $\square$

5.2. The scan. Theorem 2.2 turns the search for integral rows into 26 independent three-parameter searches, one for each Kodaira-admissible pair (ρ, δ_∞) with $r_i = j_i/M \in [-1, 2]$, namely $(M; j_1, j_2)$ running over $(1; 0, 0), (2; 1, 1), (3; 1, 2), (4; 1, 3), (6; 1, 5), (3; 1, 1), (3; 2, 2), (1; 0, 1), (1; 1, 1), (2; 1, 3), (2; -1, 1), (1; -1, 1), (3; 0, 2), (3; 1, 3), (3; -1, 1), (4; -1, 1), (6; 1, 3), (6; 3, 5), (6; -1, 1), (12; 1, 7), (12; 5, 11), (2; 3, 3), (1; 1, 2), (3; 2, 4), (4; 3, 5), (6; 5, 7)$ in the notation $Q(n) = C(Mn - j_1)(Mn - j_2)$ of Theorem 2.2. Inside a class the free integers are A , the accessory parameter $B = u_1$, and C . The scan sweeps $1 \leq A \leq 3000$ (with the divisibility $2M \mid A(2M - j_1 - j_2)$), $1 \leq |B| \leq 150$, $1 \leq |C| \leq \min(4000, 12000/M^2)$, plus a separate $A = 0$ pass, testing integrality division-free on $U_n = u_n(n!)^2$,

$$U_{n+1} = P(n)U_n - n^2Q(n)U_{n-1}, \quad u_n \in \mathbb{Z} \iff v_p(U_n) \geq 2v_p(n!),$$

prime by prime modulo p^K for every $p \leq 30$. Survivors are re-verified with exact rational arithmetic to $n = 300$; the sharp denominator exponent k of the companion ($b_0 = 0, b_1 = 1$) is measured to $n = 60$. The sweep is 1 270 065 hits in about 75 minutes on twelve cores.

5.3. Two filters. The Casoratian satisfies $u_n b_{n+1} - u_{n+1} b_n = \pm \prod_{m \leq n} Q(m)/((m+1)!)^2$, so it vanishes identically as soon as $Q(n_0) = 0$ for an integer $n_0 \geq 1$ (i.e. $M \mid j_i$ with $j_i/M \geq 1$); the companion is then a rational multiple of the row, the Apéry limit is rational, and the extension class is split. And $a^2 = 4d$ gives $\lambda_1 = \lambda_2$, hence no second growth rate. Both are applied before any row is called interesting. The Casoratian filter is not cosmetic: in one early run it reduced 25 390 raw hits to 248, and without it the entire top of the raw ranking consists of degeneracies.

5.4. Rigid classes and family classes. The scan output splits sharply, and the split is Herfurtner's own. Classes attached to rigid configurations yield a handful of rows each (10–140 hits); two classes yield tens of thousands, namely $(2; -1, 1)$, i.e. $(\rho; \delta_\infty) = (0; 1)$, with 28 143 hits, and $(2; 1, 3)$, i.e. $(1; 1)$, with 34 430 — precisely the classes in which the fibre at ∞ has half-integral exponents (an I_m^*), the

#	class	$P(n)$	$Q(n)$	λ_1	λ_2	k	score	$\mathcal{J}$ / name
1	(1; 0, 0)	$9n^2+9n+3$	$27n^2$	5.196	cplx	2	—	yes (1, 12), Zagier B
2	(2; 1, 1)	$56n^2+28n+6$	$324(2n-1)^2$	36	cplx	2	—	no, $\sqrt{AZ(7, 3, 81)}$
3	(2; 1, 1)	$88n^2+44n+10$	$500(2n-1)^2$	44.72	cplx	2	—	yes (1, 6), $\sqrt{AZ(11, 5, 125)}$
4	(3; 1, 1)	$117n^2+78n+21$	$441(3n-1)^2$	63	cplx	2	—	yes (1, 8), NEW $I_1 I_7 III III$
5	(3; 1, 2)	$40n^2+20n+4$	$48(3n-1)(3n-2)$	20.78	cplx	2	—	no, new
6	(4; 1, 3)	$72n^2+36n+6$	$108(4n-1)(4n-3)$	41.57	cplx	2	—	yes (3, 3), NEW $I_3 III III III$
7	(1; 0, 0)	$11n^2+11n+3$	$-n^2$	11.090	-0.0902	2	+0.406	yes (1, 12), Zagier D
8	(2; 1, 1)	$136n^2+68n+10$	$4(2n-1)^2$	135.88	0.1177	2	+0.139	no, Beukers = $\sqrt{Apéry}$
9	(2; 1, 1)	$24n^2+12n+2$	$4(2n-1)^2$	23.314	0.6863	2	-1.624	no, $\sqrt{T}$
10	(4; 1, 3)	$88n^2+44n+6$	$-4(4n-1)(4n-3)$	88.721	-0.7214	2	-1.673	no, level-5 Fricke
11	(1; 0, 0)	$7n^2+7n+2$	$-8n^2$	8	-1	2	-2.000	yes (1, 12), Zagier A
12	(1; 0, 0)	$10n^2+10n+3$	$9n^2$	9	1	2	-2.000	yes (1, 12), Zagier C
13	(3; 1, 2)	$26n^2+13n+2$	$-3(3n-1)(3n-2)$	27	-1	2	-2.000	no, $\sqrt{s_7}$
14	(3; 1, 1)	$180n^2+120n+24$	$-72(3n-1)^2$	183.53	-3.531	2	-3.262	no ($\mathcal{I} = -50$), new
15	(1; 0, 0)	$12n^2+12n+4$	$32n^2$	8	4	2	-3.386	yes (1, 12), Zagier E
16	(2; 1, 1)	$20n^2+10n+2$	$16(2n-1)^2$	16	4	2	-3.386	no, $\sqrt{Domb}$
17	(4; 1, 3)	$28n^2+14n+2$	$-8(4n-1)(4n-3)$	32	-4	2	-3.386	no, new
18	(2; 1, 1)	$72n^2+36n+6$	$-108(2n-1)^2$	77.569	-5.569	2	-3.717	yes (3, 6), $\sqrt{AZ(9, 3, -27)}$
19	(1; 0, 0)	$17n^2+17n+6$	$72n^2$	9	8	2	-4.079	yes (1, 12), Zagier F
20	(4; 1, 3)	$80n^2+40n+6$	$36(4n-1)(4n-3)$	72	8	2	-4.079	no, new
21–22	(2; 1, 3)	± 4	$-64(2n-1)(2n-3)$	16	-16	2	-4.773	no, $A = 0$
23	(4; 1, 3)	$48n^2+24n+4$	$32(4n-1)(4n-3)$	32	16	2	-4.773	no, new
24–25	(3; 2, 4)	± 3	$-81(3n-2)(3n-4)$	27	-27	2	-5.296	no, $A = 0$
26	(4; 1, 3)	$68n^2+34n+6$	$72(4n-1)(4n-3)$	36	32	2	-5.466	no, new
27–28	(4; 3, 5)	± 4	$-256(4n-3)(4n-5)$	64	-64	2	-6.159	no, $A = 0$

TABLE 2. All primitive, Casoratian-non-degenerate, non-double-root integral rows produced by the scan in the classes attached to rigid configurations, sorted by score. “cplx” means complex-conjugate characteristic roots, hence no archimedean Apéry limit. [VERIFIED: integrality exact to $n = 300$; k sharp to $n = 60$; $\mathcal{J}$ -fits exact]

second additionally having an apparent singularity (I_0^*) at t_1, t_2 . Those are exactly the Kodaira types that put a configuration into Herfurtnier’s one-parameter block. For $(2; -1, 1)$ the family is explicit: $B = \frac{A}{4}$, $P(n) = \frac{A}{4}(2n+1)^2$, $Q(n) = C(2n-1)(2n+1)$, integral for $A \in 4\mathbb{Z}$ and C in an arithmetic progression — one parameter after $t \mapsto t/\mu$, matching the one modulus of Herfurtnier’s $I_1 I_1 I_1 I_3^*$ and $I_1 I_1 I_2 I_2^*$ families; its $C = 4$ sub-family is the classical Legendre–Padé row with Apéry limit $-\frac{1}{12} \log \frac{x+1}{x-1}$ at $A = 8x$, which has $k = 1$, score $\rightarrow \infty$, and proves nothing new. *This is the structural dichotomy of the classification:* a rigid configuration supports at most finitely many integral rows (Theorem 3.2 pins a^2/d and the surface then determines the accessory parameter), a one-parameter family supports an infinite family of them, and Zagier’s finiteness statement is a statement about the rigid block.

5.5. **The rows.** Table 2 is the answer in the rigid block. Four observations.

- (1) *Every second-order row of the known census appears*, with its known invariants: Zagier’s six, the six Almkvist–Zudilin square roots, Cooper’s $\sqrt{s_7}$, the level-5 Fricke row. And $k = 2$ for every row in the table without exception.
- (2) *Two rows are new and are elliptic surfaces* (rows 4 and 6); see §5.6.
- (3) Six further rows are new but fail the $\mathcal{J}$ -test (rows 5, 14, 17, 20, 23, 26); like Beukers’ row they live on Atkin–Lehner quotients.
- (4) *No new row has a positive score.*

5.6. **The two new elliptic-surface rows.** Both are invisible to Zagier’s normalisation ($\rho \neq 0$) and to the root-row normalisation, which is why no earlier scan found them. They are found only by running the exponent classification first and scanning each Kodaira-admissible class in turn.

Proposition 5.1. [VERIFIED: integrality $n \leq 300$ exact; $\mathcal{J}$ -fit exact; uniformisation to q^{90} resp. w^{45}]

$$(a) \quad (n+1)^2 u_{n+1} = (117n^2 + 78n + 21)u_n - 441(3n-1)^2 u_{n-1}, \quad u_0 = 1,$$

$$u_n = 1, 21, 693, 23940, 734643, 13697019, \dots \in \mathbb{Z},$$

class $(\rho; \delta_\infty) = (\frac{1}{3}; 0)$, $\mathcal{I} = \frac{169}{49}$, $\deg \mathcal{J} = 8$, $h = 1$: the surface $I_1 I_7 II II$ (Herfurtnier #30), projective monodromy $\Gamma_0(7)$. Uniformisation:

$$t = -\frac{1}{9} \left(\frac{\eta(7\tau)}{\eta(\tau)} \right)^4, \quad A(t) (1 - 117t + 3969t^2)^{1/3} = \theta_{-7}(\tau),$$

with θ_{-7} the weight-one theta series of $\mathbb{Q}(\sqrt{-7})$, the unique form in $M_1(\Gamma_0(7), \chi_{-7})$; equivalently $\theta_q t/t = \theta_{-7}^2 \in M_2(\Gamma_0(7))$. Moreover

$$\frac{t \theta_{-7}(\tau)^3}{1 - 117t + 3969t^2} = -\frac{1}{9} \eta(\tau)^3 \eta(7\tau)^3,$$

the weight-three newform 7.3.b.a. Since $\lambda^2 - 117\lambda + 3969$ has discriminant -2187 , $|\lambda_1| = |\lambda_2| = 63$ and there is no archimedean Apéry limit; $k = 2$. The row is integral only after the scaling $\lambda = 3$.

$$(b) \quad (n+1)^2 u_{n+1} = (72n^2 + 36n + 6)u_n - 108(4n-1)(4n-3)u_{n-1}, \quad u_0 = 1,$$

$$u_n = 1, 6, 90, 1140, -5850, -1265004, \dots \in \mathbb{Z},$$

class $(\frac{1}{2}; \frac{1}{2})$, $\mathcal{I} = 3$, $\deg \mathcal{J} = 3$, $h = 3$: the surface $I_3 III III III$ (#45, $\mathcal{J} = X^3/Y^3$), projective monodromy the index-three congruence group of level three, the kernel of $\mathrm{PSL}_2(\mathbb{Z}) \rightarrow \mathbb{Z}/3$. Uniformisation:

$$j = \frac{(24t-1)^3}{8t^3}, \quad 3\theta_q t/t = A(t)^2 (1 - 72t + 1728t^2)^{1/2},$$

with $t = -\frac{1}{2}w - 6w^2 - 72w^3 - 740w^4 - \dots$ a series in $w = q^{1/3}$. $\lambda^2 - 72\lambda + 1728$ has discriminant -1728 , $|\lambda_i| = 41.57$, no archimedean limit; $k = 2$; integral only after $\lambda = 6$.

The two behave differently, and the difference is the gauge exponent. At #30 it is $\frac{1}{3}$ and the two II fibres are order-3 elliptic points, so $R^{1/3}$ is single-valued on $\mathbb{H}$ and $A \cdot R^{1/3}$ is a genuine holomorphic modular form; at #45 it is $\frac{1}{4}$ while the three III fibres are order-2 points, so $R^{1/4}$ is not single-valued and the row's " F " is a square root of a weight-two object — the quadratic twist that Theorem 2.3 allows. In neither case is the Eichler source $\Phi = D^2(B/A)$ a weight-three modular form on the relevant group: for #30 its expansion $-\frac{1}{9}q - \frac{4}{27}q^2 - \frac{17}{81}q^3 + \dots$ has unbounded 3-power denominators and the fit against $M_3(\Gamma_0(7), \chi_{-7})$ fails from q^4 on. Since the characteristic roots are complex conjugate the natural constant is not an Apéry limit but a *fold constant*, the connection coefficient at the complex singular point t_c ; both folds sit at classical CM points ($j(\tau_*) = 1728$, discriminant -4 , for #45, as it must be with the III fibres over $j = 1728$; $\tau_* \sim \rho$, discriminant -3 , for #30), so Chowla–Selberg periods are legitimate candidates alongside L -values. For #45 the identification succeeds.

Proposition 5.2 (a Chowla–Selberg period). [NUMERICAL: ≥ 245 digits; agreement 1.0×10^{-249}] For the row #45,

$$\mathrm{Im} \xi = -\frac{\Gamma(1/4)^4}{2^{7/2} 3^{11/4} \pi} = -\frac{\varpi^2}{\sqrt{2} 3^{11/4}},$$

$\varpi = \Gamma(1/4)^2 / (2\sqrt{2}\pi)$ the lemniscate constant, the Chowla–Selberg period of discriminant -4 ; equivalently $4 \cdot 3^{11} (\mathrm{Im} \xi)^4 = \varpi^8$, and `algdep` returns $2^{14} 3^{11} x^4 - 1$ for $\mathrm{Im} \xi / (\Gamma(1/4)^4 \pi^{-1})$ at degree 4, height 2×10^9 .

This is the first constant in this census that is a Chowla–Selberg period rather than a critical L -value — a direct consequence of the source not being modular. The corresponding constants for #30 resist identification against every basis tried (Chowla–Selberg monomials, Dirichlet and newform L -values, `algdep` with algebraic coefficients to degree 18). [OPEN]

Finally, the class $(\frac{1}{2}; \frac{1}{2})$ has Fuchsian signature $(0; 2, 2, 2; \text{one cusp})$, and *two* groups of that signature occur among integral rows: the index-three subgroup of $\mathrm{PSL}_2(\mathbb{Z})$ above (row 6) and $\Gamma_0(5) + 5$, of the

same covolume but not conjugate into $\mathrm{PSL}_2(\mathbb{Z})$, carrying the level-5 Fricke row (row 10). The $\mathcal{J}$ -map test is exactly what separates them.

5.7. The classification theorem. Theorem D in full; parts (i)–(ii) are Theorems A and B and are not repeated.

Theorem 5.3 (classification of second-order rows). *Let (1) have $u_n \in \mathbb{Z}$ for all n (after a λ^n rescaling), $\deg P, \deg Q \leq 2$, $d \neq 0$, $a^2 \neq 4d$, $Q(n) \neq 0$ for $n \geq 1$.*

- (iii) [VERIFIED: all 26 Kodaira-admissible classes, $1 \leq A \leq 3000$, $|B| \leq 150$, $|C| \leq \min(4000, 12000/M^2)$, integrality prime-tested to $n = 30$ and re-verified exactly to $n = 300$] *In the classes attached to rigid configurations the complete list of primitive, Casoratian-non-degenerate rows is the 28 rows of Table 2. Exactly ten have an elliptic-surface local system: Zagier’s six ($\deg \mathcal{J} = 12$; Beauville’s $I_1 I_2 I_3 I_6$ three times, $I_1 I_1 I_1 I_9$, $I_1 I_1 I_5 I_5$, $I_1 I_1 I_2 I_8$), the two root rows of Theorem 4.5 ($\deg \mathcal{J} = 6$), and the two new rows of Proposition 5.1 ($\deg \mathcal{J} = 8$ and 3). The other eighteen have projective monodromy commensurable with, but not conjugate into, $\mathrm{PSL}_2(\mathbb{Z})$.*
- (iv) [VERIFIED: same box] *In the classes attached to Herfurtners’ one-parameter families the integral rows form infinite families; the cleanest is $(\rho; \delta_\infty) = (0; 1)$, $P(n) = \frac{A}{4}(2n+1)^2$, $Q(n) = C(2n-1)(2n+1)$, integral for $A \in 4\mathbb{Z}$ and C in an arithmetic progression — one parameter after $t \mapsto t/\mu$, matching the one modulus of Herfurtners’ $I_1 I_1 I_1 I_3^*$ / $I_1 I_1 I_2 I_2^*$ families.*

Corollary 5.4 (the only positive scores). [VERIFIED: complete pass over all 1 270 065 scan hits: a positive score needs $|\lambda_2| < e^{-k} \leq e^{-1}$, a condition on (A, D) alone] *Of the 69 434 hits in Casoratian-non-degenerate classes, 1 374 have $|\lambda_2| < 1$ with real characteristic roots, and exactly two have a positive score with $k \geq 2$:*

$$\text{Apéry's } \zeta(2) \text{ row } (11, 3, -1) \text{ at } +0.40606, \quad \text{Beukers' row } (136, 10, 4) \text{ at } +0.13920.$$

Every other positive score in the scan has $k = 1$ and belongs to the classical Legendre–Padé log family (top of that ranking: $A = 600$, score $+2.5115$, $k = 1$).

Remark 5.5 (what the scan does not settle). The eleven Herfurtners one-parameter families with a moving $\mathcal{J}$ are not excluded by Theorem 3.2 — their cross-ratio is free — only by the scan; and a box is a box. [OPEN] The proved part of the classification is entirely local (Lemma 2.1, Theorems 2.2, 2.3, 3.2, Proposition 4.2); the global input is Herfurtners’ theorem, and the finiteness of the answer is the finiteness of his list together with the rationality of a^2/d .

5.8. What an unconditional classification would need. Theorem A is conditional on the geometric hypothesis; Theorem 5.3 replaces it by a box. Neither is the statement one wants, which is Zagier’s conjecture as quoted in §1: here his $P(t)$ is our $R(t) = 1 - at + dt^2$, so “non-degenerate” is exactly $d \neq 0$, $a^2 \neq 4d$. This is the rank-two, four-singular-point instance of the Bombieri–Dwork expectation [5] that a globally nilpotent Fuchsian operator is of geometric origin: an integral row’s solution is a G -function, so by the Chudnovsky theorem [9] its minimal operator is globally nilpotent, and what is missing is the passage from that to a family of varieties. The converse reduction is proved here.

Proposition 5.6. [PROVED] (modulo the classification inputs already used) *Suppose an integral row (1) with $d \neq 0$, $a^2 \neq 4d$ has a modular parametrisation: t a Hauptmodul of a genus-zero Γ commensurable with $\mathrm{PSL}_2(\mathbb{Z})$, the solution the t -expansion of a weight-one form on Γ . Then by Lemma 7.1 the singular points are the special points of Γ , so Γ has exactly four of them; by Theorem 7.3 its covolume index is ≤ 12 and, if $\Gamma \subseteq \mathrm{PSL}_2(\mathbb{Z})$, it is one of 28 enumerated groups; and Theorems A and 7.4 then place the row on the list. Hence Zagier’s conjecture for this class is equivalent to the statement that the projective monodromy group of an integral three-term row is commensurable with $\mathrm{PSL}_2(\mathbb{Z})$.*

Why that is hard was set out in §1: rank two with four singular points is not rigid in Katz’s sense [11], the accessory parameter being the modulus; the implication “globally nilpotent $\Rightarrow$ hypergeometric pullback” may not be used as a shortcut, Krammer’s Lamé equation [13, 15] being the standing warning; and the strongest partial result, Beukers’ p -adic analysis of Dwork’s accessory parameter problem in exactly this configuration [4], does not close it.

What the computations supply instead is a census, and it is consistent: over every box searched, every integral row is (a) one of the ten elliptic-surface rows of Theorem 5.3(iii); or (b) a row on an Atkin–Lehner quotient — Beukers’/ $\sqrt{\text{Apéry}}$, $\sqrt{T}$, $\sqrt{\text{Domb}}$, $\sqrt{\text{AZ}(7,3,81)}$, $\sqrt{s_7}$, the level-5 Fricke row, six further new rows and the $A = 0$ rows, all still *modular* in Zagier’s sense, the universal curve simply failing to descend to their base; or (c) a member of one of the two classical Padé families — the Legendre–Padé log family, which occupies the classes attached to Herfurtners’ one-parameter I_m^*/I_0^* families, and the arctan family, which Theorem B(ii) excludes from the elliptic-surface world and whose canonical coordinate is not even integral (lcm of denominators over $n \leq 58$ equal to $2^{11}5^67^411^213^2 \cdot 17 \cdot 19 \cdot 23$, new primes entering indefinitely, so $t(q)$ is not algebraic by Eisenstein’s theorem [VERIFIED: $n \leq 58$]). No row in any box is unaccounted for; but Proposition 5.6 is where the argument stops. [OPEN]

6. CUSP MOVES AND RIGIDITY

Table 1 shows three of Zagier’s rows on one surface. That is not a coincidence of cross-ratios but a group action, and the action has a sharp arithmetic consequence.

Theorem 6.1 (the general cusp move). [PROVED] (*symbolically*) *Let y solve $Ly = 0$ for a row (P, Q) with characteristic roots λ, μ and Q -roots r_1, r_2 . Put*

$$s = \frac{t}{1 - \lambda t} \quad \left(\text{so } t = \frac{s}{1 + \lambda s} \right), \quad z(s) = (1 - \lambda t)^\alpha y(t).$$

Then z is again the analytic solution of a three-term row with coefficients of degree ≤ 2 if and only if

$$(\alpha - 1 + r_1)(\alpha - 1 + r_2) = 0, \quad \text{i.e.} \quad \alpha \in \{1 - r_1, 1 - r_2\},$$

i.e. exactly when α is one of the two local exponents of L at $t = \infty$. With $\alpha = 1 - r_1$ the new row is

$$\begin{aligned} P^\sharp(n) &= (\mu - 2\lambda)n^2 + (p_1 - 3\lambda + 2\lambda r_1)n + (p_0 - \lambda + \lambda r_1), \\ Q^\sharp(n) &= \lambda(\lambda - \mu)n^2 + \lambda(\lambda + \mu - p_1 - 2\lambda r_1 + \mu r_1 - \mu r_2)n \\ &\quad + \lambda(\lambda r_1^2 - \lambda r_1 + \mu r_1 r_2 - \mu r_1 + p_1 r_1), \end{aligned}$$

with characteristic roots $-\lambda$ and $\mu - \lambda$, and the two operators are related by $L^\sharp z = (1 - \lambda t)^{1-\alpha} Ly$.

Proof sketch. Substituting $t = s/(1 + \lambda s)$ and $y = (1 + \lambda s)^\alpha z$ into Ly and dividing by $(1 + \lambda s)^{\alpha-1}$ makes the coefficients of z'' and z' polynomial of degree 2 unconditionally, while the coefficient of z is polynomial with remainder

$$\frac{\alpha^2 \lambda \mu - 2\alpha \lambda \mu - \alpha q_1 + \lambda \mu + q_0 + q_1}{\lambda^2} = \frac{d}{\lambda^2} (\alpha - 1 + r_1)(\alpha - 1 + r_2)$$

after $q_1 = -d(r_1 + r_2)$, $q_0 = dr_1 r_2$. Setting this to zero is the stated condition, and the boxed coefficients are then read off. The computation is a finite identity in the Weyl algebra and was carried out symbolically. $\square$

The reading is transparent: the move sends $t = \infty$ to $s = -1/\lambda$, where a three-term row needs local exponents $(0, \rho^\sharp)$; the gauge $(1 - \lambda t)^\alpha$ shifts the exponents there by $-\alpha$, and the two admissible choices are the two exponents $1 - r_i$. So the move is the exponent bookkeeping of Theorem 2.2 turned into a group action. In Zagier’s class ($Q = dn^2$, $r_1 = r_2 = 0$, $\alpha = 1$) it specialises to $(a, b, d) \mapsto (\mu - 2\lambda, b - \lambda, \lambda^2 - \lambda\mu)$ and, on coefficients, to $u_n^\sharp = \sum_m \binom{n+\alpha-1}{n-m} (-\lambda)^{n-m} u_m$, the signed binomial transform; for $\alpha = 1$, $\lambda \in \mathbb{Z}$ integrality is automatic, while for $\alpha \notin \mathbb{Z}$ the binomials carry

denominators at the primes dividing $\text{den}(\alpha)$ and a rescaling $u_n \mapsto c^n u_n$ with $c \in \{1, 2, 4, 9\}$ restores it in every case measured.

Theorem 6.2 (the companion). [PROVED]; [VERIFIED: all 22 available (row, λ , α) cases, exactly to $n = 40$] *In the situation of Theorem 6.1, let $\hat{B}$ solve $L\hat{B} = t(1 - \lambda t)^{-\alpha}$ with $\hat{b}_0 = 0$. Then the companion of the moved row is $B^\sharp = (1 - \lambda t)^\alpha \hat{B}$. In particular the naive guess $B^\sharp = (1 - \lambda t)^\alpha B$ is false: what is replaced is the inhomogeneity, not the companion.*

Proof. By Theorem 6.1, $L^\sharp[(1 - \lambda t)^\alpha X] = (1 - \lambda t)^{1-\alpha} L X$ for any X . With $X = \hat{B}$, $L^\sharp B^\sharp = (1 - \lambda t)^{1-\alpha} \cdot t(1 - \lambda t)^{-\alpha} = t/(1 - \lambda t) = s$, which is the defining inhomogeneity of the moved row's companion; and $B^\sharp = s + O(s^2)$. Uniqueness of the analytic solution with $b_0 = 0$ finishes. $\square$

Consequently the Apéry limit is *not* preserved by a cusp move: two things change at once, the dominant singularity and the second solution. In modular terms, with $\Phi = F Dt$ the weight-three source and $\Theta = D^{-2}\Phi$, $B^\sharp = F^\sharp D^{-2}(\Phi/(1 - \lambda t))$; multiplication by the modular unit $(1 - \lambda t)^{-1}$ is not Hecke-equivariant, so the source can jump between Hecke eigenspaces.

Theorem 6.3 (orbits and rigidity). [PROVED] *In the coordinate $v = 1/t$ the four singular points are $v = \infty$ (MUM), 0 , λ , μ , and the cusp move by λ is the translation $v \mapsto v - \lambda$. Hence the orbit consists of exactly three placements — which of $\{0, \lambda, \mu\}$ goes to $v = 0$ — each carrying at most four gauges, so at most 12 rows modulo $u_n \mapsto c^n u_n$; and the placement at $v = v_j$ has $|\lambda_2| = \min_{i \neq j} |v_i - v_j|$. Moreover (Theorem B(iii)), if $a^2 - 4d$ is a square then all three gaps $|\lambda|, |\mu|, |\lambda - \mu|$ are positive integers, so $|\lambda_2| \geq 1$ and $\text{score} \leq -k \leq -1$ at every placement.*

Corollary 6.4. [PROVED] *No rational cusp-move orbit contains a positive-score row; conversely a positive-score row has irrational characteristic roots, so its only other equally-good placement is its Galois conjugate over $K = \mathbb{Q}(\sqrt{a^2 - 4d})$.*

Measured orbit sizes in the corpus of 49 distinct rows are 2, 3, 8, 12; the **A**, **C**, **F** orbits coincide, all members are integral to $n = 200$ after a rescaling $c \in \{1, 2, 4, 9\}$, $k \in \{1, 2\}$ throughout and sharp, the best score in a rational orbit is exactly -2 , and the only corpus rows with $|\lambda_2| < 1$ — Apéry's $\zeta(2)$ row (0.09017), Beukers' (0.11775), $\sqrt{T}$ (0.68629), $\Gamma_0(5) + 5$ (0.72136) — all have irrational roots. [VERIFIED: 49 rows, 8 orbits, $n \leq 200$] The conjugate placements exist over K with the *same* score to the last digit but approximate a different real number: 0.32781709442... against $\zeta(2)/5 = 0.32898681337...$, and 0.10021505333... against 0.10018744923... [NUMERICAL: ≥ 231 digits]

6.1. One surface, two extension classes. With $t_C = \eta_1^4 \eta_6^8 / (\eta_2^8 \eta_3^4)$, $F_C = \eta_2^6 \eta_3 / (\eta_1^3 \eta_6^2)$ and **F'** denoting **F** in the level-6 normalisation $u_n^{\mathbf{F}'} = (-1)^n u_n^{\mathbf{F}}$,

$$t_A = \frac{t_C}{1 - t_C}, \quad t_{\mathbf{F}'} = \frac{t_C}{1 - 9t_C}, \quad F_A = (1 - t_C)F_C, \quad F_{\mathbf{F}'} = (1 - 9t_C)F_C$$

[VERIFIED: exact to $O(q^{241})$]. The cusps of $\Gamma_0(6)$ are $\infty, 1/2, 1/3, 0$ of widths 1, 3, 2, 6, carrying I_1, I_3, I_2, I_6 ; t_C vanishes at ∞ , has its pole at $1/2$ and takes the values $1, \frac{1}{9}$ at $1/3, 0$, which is why the three substitutions put I_3, I_2, I_6 at $t = \infty$. The relating map is *not* an Atkin–Lehner involution: $\{1, W_2, W_3, W_6\}$ acts simply transitively on the cusps, so no nontrivial element fixes the MUM cusp.

The surface fixes the pure local system; the choice of MUM cusp fixes which Eisenstein class degenerates there. With $\mathcal{S} = E_{3, \chi_{-3}, 1}$, $\mathcal{T} = E_{3, 1, \chi_{-3}}$ and $(V_k f)(\tau) = f(k\tau)$, one has $\Phi_C = (1 - 8V_2)\mathcal{S}$, $\Phi_{\mathbf{F}'} = (1 + V_2)\mathcal{S}$, $\Phi_A = (1 - V_2)\mathcal{T}$ [VERIFIED: exact to $O(q^{260})$]. The four series $\mathcal{S}, V_2\mathcal{S}, \mathcal{T}, V_2\mathcal{T}$ are independent, so $\{\mathcal{S}, V_2\mathcal{S}\}$ and $\{\mathcal{T}, V_2\mathcal{T}\}$ span two different planes of $\mathcal{M}_3(\Gamma_0(6), \chi_{-3})$; the cusp move multiplies Φ by the modular unit $(1 - \lambda t)^{-1}$ and is not Hecke-equivariant, landing in the $\mathcal{S}$ -plane for $\lambda = 9$ and in the $\mathcal{T}$ -plane for $\lambda = 1$. Hence *one surface, two extension classes*: $\xi^{\mathbf{C}} = \frac{1}{2}L(2, \chi_{-3})$, $\xi^{\mathbf{F}'} = -\frac{5}{8}L(2, \chi_{-3})$, so $\xi^{\mathbf{F}'} / \xi^{\mathbf{C}} = -\frac{5}{4}$ exactly, while $\xi^{\mathbf{A}} = \frac{1}{4}\zeta(2)$ and $\xi^{\mathbf{A}} / \xi^{\mathbf{C}}$ has no rational approximation of height $\leq 10^{12}$ beyond the random level. [NUMERICAL: ≥ 80 digits]

Corollary 6.5 (the 3-adic relation). *Inside the $\mathcal{S}$ -plane, $\Theta_{\mathbf{F}'} = -\frac{5}{4}\Theta_{\mathbf{C}} + \frac{9}{4}\Theta_{\Psi_0}$ with $\Psi_0 = (1 - 4V_2)\mathcal{S}$ the class of vanishing Mellin factor ([VERIFIED: $O(q^{260})$]); multiplying by $F_{\mathbf{F}'} = (1 - 9t_{\mathbf{C}})F_{\mathbf{C}}$ gives*

$$B^{\mathbf{F}'} + \frac{5}{4}\xi A^{\mathbf{F}'} = (1 - 9t_{\mathbf{C}})\left(-\frac{5}{4}R_{\xi}^{\mathbf{C}} + \frac{9}{4}\Xi\right), \quad t_{\mathbf{C}} = \frac{t_{\mathbf{F}'}}{1 + 9t_{\mathbf{F}'}}.$$

Since $x \mapsto x/(1 + 9x)$ is an isometric automorphism of the disc $\{|x|_3 \leq 3^\delta\}$, $0 < \delta < 2$, and $1 \pm 9x$ are units of norm 1 there, $\xi_3^{\mathbf{F}} = \frac{5}{4}\xi_3^{\mathbf{C}}$ — conditionally on overconvergence of $R_{\xi}^{\mathbf{C}}$ and Ξ on such a disc and on $v_3(a_n^{\mathbf{F}}) = O(\log n)$. [NUMERICAL: $v_3(4\xi_3^{\mathbf{F}} - 5\xi_3^{\mathbf{C}}) = 1586$ at $N = 800$, against Cauchy precision 1581]

The move transports nothing to $\mathbf{A}$, and cannot: $\Theta_{\mathbf{C}}, \Theta_{\Psi_0}, \Theta_{\mathbf{A}}$ have rank 3, so no such identity exists; $\mathbf{A}$ has $\sigma_3 = v_3(d) = 0$ and no 3-adic Apéry limit at all; and at its own alignment prime $\xi_2^{\mathbf{A}} = 0$. The failure is structural, not a gap.

6.2. The cusp move can change the source's Hecke type. The $\sqrt{\text{Domb}}$ orbit (size 8) contains two members with genuinely mixed limits: $(20, 10, 64)$ with $\xi = \frac{1}{16}(15L(2, \chi_{-3}) - 6\zeta(2))$ and $(20, 4, 64)$ with $\xi = \frac{1}{32}(6\zeta(2) + 15L(2, \chi_{-3}))$, both integral to $n = 200$ with $k = 2$ sharp, $\lambda_1 = 16$, $\lambda_2 = 4$. [NUMERICAL: agreement 2.8×10^{-124} resp. 2.7×10^{-118} at $n = 200$ against predicted truncation 3.9×10^{-121}] Their parent carries the *cuspidal* value $L(f_{12}, 2)$: the cusp move has pushed the weight-three source out of the cuspidal part into the Eisenstein part of $M_3(\Gamma_0(12), \chi_{-3})$, which Theorem 6.2 explains. These are the first rows in this census whose single Apéry limit is a nontrivial $\mathbb{Q}$ -combination of $\zeta(2)$ and $L(2, \chi_{-3})$, with independent coefficient vectors $(-6, 15)$ and $(6, 15)$ — the shape a Nesterenko argument for $\dim\langle 1, \zeta(2), L(2, \chi_{-3}) \rangle$ wants. It does not get near: here $\sigma = 2 + \log 16$ and $\delta = -2 - \log 4$, a deficit of 3.386 nats per step to $\dim \geq 2$, and from the better pair $(\mathbf{A}, \mathbf{C})$ the deficit to $\dim \geq 3$ — the theorem of Calegari–Dimitrov–Tang [7] — is 10.394 nats per step.

7. NON-CONGRUENCE HOSTS

The classification above is organised by the surface. There is a second organisation, by the group, and it produces the same finiteness from Gauss–Bonnet rather than from Riemann–Hurwitz.

Let Γ be a genus-zero Fuchsian group commensurable with $\text{PSL}_2(\mathbb{Z})$ with a distinguished cusp, t a Hauptmodul normalised $t = q_h + O(q_h^2)$ with integral coefficients, and F a weight-two form on Γ with $F = 1 + O(q)$ and divisor supported on the cusps (an eta quotient, say). Put $g = \sqrt{F}$ and $a_n = \lambda^n[t^n]g$. Then g is a weight-one form on an index- ≤ 2 subgroup, a_n satisfies a second-order row, and $4^n[t^n]\sqrt{G} \in \mathbb{Z}$ for any $G \in 1 + x\mathbb{Z}[[x]]$ — sharply, since $4^k\binom{1/2}{k} = (-1)^{k-1}2C_{k-1}$ with C_j Catalan — so $\lambda = \max(1, 2^{2-e}) \in \{1, 2, 4\}$ where $e = \min_{n \geq 1} v_2([t^n]F)$.

Lemma 7.1. [PROVED] *The finite singular points of the weight-one operator in the t -line are exactly the values $t(P)$ for P a cusp or elliptic point of Γ with $t(P) \notin \{0, \infty\}$. They do not depend on F .*

Proof. At a non-elliptic $P \in \mathbb{H}$, t is a local isomorphism and g is holomorphic and non-vanishing, so both solutions are holomorphic and independent and $t(P)$ is ordinary. At an elliptic point of order e the local coordinate on X_Γ is $w \sim (\tau - \tau_P)^e$ and $t - t(P) \sim w$, so the local monodromy is a rotation by $2\pi/e$ and the exponent difference is $1/e$. At a cusp with t finite the second solution carries $\log(t - t(c))$. Zeros or poles of F change the exponents, not the locations. $\square$

Theorem 7.2 (the score splits). [PROVED] *With $k = 2$ as above and $|t_1| \leq |t_2| \leq \dots$ the singular values,*

$$\text{score} = \log(1/|\lambda_2|) - k = \log |t_2| - 2 - \log \lambda.$$

The archimedean factor $|t_2|$ depends only on (Γ, t) — not on F — and $\lambda \in \{1, 2, 4\}$ depends only on the 2-adic structure of F . Hence a positive score needs $|t_2| > e^2\lambda$, i.e. $|t_2| > 7.389, 14.778$ or 29.556 according as $\lambda = 1, 2, 4$.

By Calegari–Dimitrov–Tang [6], a modular form on a finite-index subgroup with algebraic Fourier coefficients and *bounded* denominators is a form on a congruence subgroup. So $\lambda > 1$ is a certificate of non-congruence; the converse direction, and the value of λ , come from the elementary binomial estimate above, not from their theorem.

Theorem 7.3 (four special points; finiteness). [PROVED] (*Gauss–Bonnet*)+[VERIFIED: exhaustive enumeration] *A three-term row forces Γ to have exactly four special points (cusps plus elliptic points), whence its covolume index in $\mathrm{PSL}_2(\mathbb{Z})$ is*

$$\mu = 6 \left(2 - \sum_i \frac{1}{e_i} \right) \leq 12,$$

with equality exactly in the four-cusp case. Inside $\mathrm{PSL}_2(\mathbb{Z})$ there are, up to conjugacy, exactly 28 genus-zero subgroups with four special points: 16 congruence and 12 non-congruence. The non-congruence ones have index 7, 9, 10 and signatures $(0; 2, 3; 2)$, $(0; 2; 3)$, $(0; 3; 3)$, four groups each. The index-12 four-cusp case returns exactly six groups, with cusp widths $[1, 1, 5, 5]$, $[1, 2, 3, 6]$, $[1, 1, 2, 8]$, $[1, 1, 1, 9]$, $[2, 2, 4, 4]$, $[3, 3, 3, 3]$, all congruence — Beauville’s six, recovered from scratch. In particular no non-congruence group has four cusps.

The enumeration is by transitive pairs $(s, t) \in S_n$ with $s^2 = t^3 = 1$, canonicalised by breadth-first relabelling, cusps read off as cycles of st , and congruence decided by Wohlfahrt’s criterion.

Theorem 7.4 (unequal exponents kill the score). [PROVED] (*modulo the normalisation $P, Q, R \in \mathbb{Z}[t]$, $P(0) = 1$, true for every row here*) *If the two finite singular points carry different exponent differences $1/e_1 \neq 1/e_2$ then $t_1, t_2 \in \mathbb{Q}$, hence $\lambda_1, \lambda_2 \in \mathbb{Z}$, so $|\lambda_2| \geq 1$ and score $\leq -k < 0$.*

This is Corollary 2.5 read on the group side, and it settles the non-congruence host question. Index 7 has signature $(0; 2, 3; 2)$ cusps): whichever of the three non-MUM special points goes to $t = \infty$, a mixed pair remains at t_1, t_2 , so all four index-7 groups are excluded outright; for index 9 and 10 the only non-mixed option is to send the elliptic point to ∞ , leaving two cusps at t_1, t_2 — the Zagier class, covered exhaustively by §5. So *a non-congruence host group contributes nothing beyond the Zagier class: the interesting non-congruence-ness is not in the group of t but in the multiplier system of $g = \sqrt{F}$.*

That is exactly Beukers’ situation. His $t = (\eta_1 \eta_6 / \eta_2 \eta_3)^{12}$ lives on the congruence group $\Gamma_0(6) + 6$; his $\sqrt{E}$ does not. With $F = \eta_2^7 \eta_3^7 / (\eta_1^5 \eta_6^5)$ Apéry’s weight-two form ($F|_2 W_6 = -F$, divisor $(1/2) + (1/3)$), $a_n = 4^n [t^n] \sqrt{F} = 1, 10, 534, 40900, 3672550, \dots$, $\lambda_{1,2} = 4(17 \pm 12\sqrt{2})$, $k = 2$, $\lambda = 4$, and the score is $+0.13920$: the criterion $\lambda e^k < |t_2|$ reads $4e^2 < (\sqrt{2} + 1)^4$, i.e. $29.5562 < 33.9706$, a margin of 1.1494. Level 6 wins because $t \circ W_6 = t$ with $t_1 t_2 = 1$ exactly, so $|t_2| = 1/|t_1|$ is as large as the fastest-growing row allows. This is Beukers’ Theorem 3 of 1987 [3]: the Apéry limit $\xi = 0.100187449229339406\dots = L(\Psi, 2)$, for the weight-three half-integral eta quotient $\Psi = \frac{1}{4}(\eta_1 \eta_6)^{9/2}(\eta_2 \eta_3)^{-3/2}$, is irrational with $\mu(\xi) \leq 50.654$.

Theorem 7.5 (uniqueness). [VERIFIED: two independent exhaustive scans; boxes stated below] *Beukers’ row is the only positive-score row whose form is non-congruence, and the only positive-score rows at all with $k \geq 2$ are it, Apéry’s $\zeta(2)$ row, and members of two infinite classical Padé families.*

- Ansatz-free recurrence scan. Integer $(\alpha, \gamma, \delta, \zeta)$ in the five classes $e \in \{2, 3, 4, 6, \infty\}$ and in a free-exponent variant, at $\alpha \leq 3000$, $|\delta| \leq 150\text{--}400$, $|\gamma| \leq 150\text{--}1000$, $|\zeta| \leq 300\text{--}2000$, with the Casoratian filter and measured k , plus a per- ρ sweep over $\rho = p/q$ with $q \leq 6$, $0 < \rho \leq 3$. Every positive score with $k \geq 2$ is Apéry’s row, Beukers’ row, a member of the Legendre–Padé log family ($\delta = 16$, $\rho = 0$, $\xi = -\frac{1}{12} \log \frac{x+1}{x-1}$ at $\alpha = 8x$), or a member of the arctan family of Corollary 2.4(b). In the square-root class $\rho = \frac{1}{2}$, Beukers’ row is the unique positive score at $\alpha \leq 3000$.
- Modular scan. All eta-quotient pairs (t, F) on $\Gamma_0(N)$, $N \leq 60$, with t of weight 0, $\mathrm{ord}_\infty t = 1$, divisor degree ≤ 4 , and F holomorphic of weight 2 with $\mathrm{ord}_\infty F = 0$: 3 036 854 pairs. Of these 739 admit a recurrence of order ≤ 4 and degree ≤ 3 (485 with $\lambda = 1$, 127 with $\lambda = 2$, 127 with

$\lambda = 4$), in 104 distinct invariant classes. Exactly two have $|\lambda_2| < 1$: Beukers' (+0.1392, $\lambda = 4$, non-congruence) and the level-8 T row (-1.6235 , $\lambda = 1$, congruence). Everything else has $|\lambda_2| \geq 1$; the next classes by $|\lambda_2|$ are 1 exactly, then 1.0635, 1.0718, 1.6180.

The modular scan has two documented blind spots — both t and F are required to be eta quotients — so the ansatz-free scan is the load-bearing one, and it is what turns up the runner-up.

Proposition 7.6 (the level-5 Fricke row). [VERIFIED: $a_n \in \mathbb{Z}$ to $n = 202$; k to $n = 200$; period to 260 digits] Let $u = (\eta_5/\eta_1)^6$, a Hauptmodul of $\Gamma_0(5)$, $t = u/(1 + 22u + 125u^2) = q - 16q^2 + 122q^3 - \dots$ the resulting Hauptmodul of $\Gamma_0(5) + 5$, and $F = E_{2,5} = \frac{1}{4}(5E_2(5\tau) - E_2(\tau))$. Then $g = \sqrt{F}$ is weight one on an index-two subgroup, $\lambda = 2$ is minimal, $a_n = 2^n[t^n]\sqrt{F} = 1, 6, 210, 10500, 615510, \dots$ satisfies $(n+1)^2 a_{n+1} = 2(44n^2 + 22n + 3)a_n + 4(4n-1)(4n-3)a_{n-1}$ with $\lambda_{1,2} = 44 \pm 20\sqrt{5} = \pm 8\varphi^{\pm 5}$, $k = 2$ sharp, non-vanishing Casoratian, and $\xi = 0.164306701064342158629932\dots$. Since $\lambda = 2 > 1$ and the source has unbounded 2-power denominators, g and Ψ live on a non-congruence group [6]: a genuine second instance of Beukers' construction, at the other perfect level. But $|t_2| = 2.7726 \ll 2e^2 = 14.778$, so the score is -1.6734 and no irrationality follows.

The comparison is exact: at level 6 the Fricke pairing is $t_1 t_2 = 1$, at level 5 it is $t_1 t_2 = -\frac{1}{4}$, and the same λ_1 -for- $|t_2|$ exchange loses a factor 4. That factor of 4 is the whole difference between the two levels. The period resisted identification against $1, \pi, \pi^2, \pi^3, \zeta(3), \log 2, \log 5, \log \varphi, \sqrt{5}$ and against $L(f, s)$, $s = 1, 2, 3$, for every weight-three newform of every level $M \leq 120$ divisible by 5. [OPEN]

One cheap diagnostic separates the Padé families from the modular rows in a single series computation: *mirror-map integrality* (§5.8). For Apéry's $\zeta(2)$ row the canonical coordinate is $q - 5q^2 + 15q^3 - 30q^4 + 40q^5 - 26q^6 - \dots$, Beukers' $\Gamma_1(5)$ Hauptmodul $q \prod (1 - q^n)^{5(\frac{n}{5})}$; for Beukers' own row it is $q - 48q^2 + 1056q^3 - \dots$, i.e. $\frac{1}{4}(\eta_1\eta_6/\eta_2\eta_3)^{12}$ in the rescaled nome; for the arctan family it is not integral at all. [VERIFIED: $n \leq 58$]

8. OUTLOOK

$K3$ and higher, and five singular points. Proposition 4.2 and Lemma 4.3 close the $K3$ worry for four singular points, but in a way worth restating precisely: an elliptic $K3$ over $\mathbb{P}_t^1$ with the row's four singular fibres is *not* excluded — in the root-row class the mixed III/III^* configuration with $\sum e = 24$ is a genuine possibility — but every such surface is a quadratic twist of a rational one with the same $\mathcal{J}$, and $\deg \mathcal{J}$ is pinned by the degree bound together with the parity, so the $\deg \leq 12$ search window was never the constraint and no new row can arise. What is not closed is a systematic pass through Shimada–Zhang's extremal elliptic $K3$ s [18] against the 26 classes. [OPEN] Five singular points means $\deg P$ or $\deg Q = 3$, i.e. a four-term recurrence or cubic coefficients; there is no analogue of Herfurtners' theorem in that generality, and the cross-ratio becomes a two-parameter modulus, so the finiteness argument of §3 does not survive. At the other end, Schmickler-Hirzebruch's classification of elliptic surfaces over $\mathbb{P}^1$ with *three* exceptional fibres [16] is the degenerate boundary: the case $d = 0$, where $t = \infty$ merges with a finite singular point and the row becomes hypergeometric — Zagier's modular solutions #11 (16, 4, 0), #14 (27, 6, 0) and #20 (64, 12, 0) in the notation of Table 1.

A gate on the higher-rank constructions. For a linear-independence argument one wants a $\mathbb{Q}$ -linear combination H of several Apéry limits whose linear form still decays — “fold-regularity”.

Lemma 8.1 (fold-regularity is codimension μ). [PROVED] Let λ_1 be the dominant characteristic root, of multiplicity μ . Near $x_1 = 1/\lambda_1$ the local solution space carries a Jordan block of size μ : solutions $y_1, y_1 \log, \dots, y_1 \log^{\mu-1}$ up to holomorphic corrections. A solution has radius of convergence $> |x_1|$ iff all μ of its coordinates on that block vanish, whereas a hypothesised $\mathbb{Q}$ -linear relation among the limits makes exactly the leading one vanish. Hence one relation implies fold-regularity iff $\mu = 1$.

For $\mu \geq 2$ the form does not decay at all — one gets polynomial convergence with $|h_n|^{1/n} \rightarrow |\lambda_1|$ — so there is nothing for a holonomy bound, or for Beukers' criterion, to act on. Verified sharply: a

row with $\mu = 1$ folds and its form decays at rate λ_2 exactly, while Sym^3 of Zagier $\mathbf{E}$ ($\mu = 3$) and a Sym^2 with $\mu = 2$ do not fold at all. [VERIFIED: n up to 10^6] Symmetric powers and Cauchy squares of a single row therefore raise μ and are the wrong place to look; genuinely independent rows on one surface (§6) are the right one.

What the classification says for irrationality proofs. The score $\log(1/|\lambda_2|) - k$ is not a property of the sequence. By Theorem 7.2 it splits as $\log|t_2| - 2 - \log \lambda$, and neither factor depends on the accessory parameter or on the period: $|t_2|$ is fixed by the surface together with the choice of which fibre sits at ∞ , and λ by the 2-adic structure of the weight-two form. The score is therefore a *motivic* invariant — of the geometry plus the placement — and the classification computes it once and for all. Combined with Corollary 6.4 (a positive score forces irrational characteristic roots, hence a real quadratic field, hence no second rational placement), Theorem 7.3 (only finitely many host geometries, all enumerated) and Corollary 5.4 and Theorem 7.5 (over every box, exactly two rows clear zero, and both were known), the honest conclusion is that the second-order world is closed. Its two positive-score rows are not improvable by rearrangement inside the list: the cusp move produces no new rational data (Theorem 6.3), merging two placements gives $\mu \leq 1 + \min_i \sigma_i/\delta_i$, i.e. the better of the two with no convex-combination gain, and the conjugate placements approximate different numbers. A new irrationality theorem of this shape must come from outside the second-order box — from a higher-rank host, or from a bound of Bost–Charles/Calegari–Dimitrov–Tang type applied to a whole function module rather than to a single linear form.

9. REPRODUCIBILITY

All computations are reproducible from the project notes and scripts [21]; PARI/GP work uses exact rational (or $\mathbb{Q}(y)/(y^2 - ay + d)$) arithmetic except where a real or p -adic precision is quoted. Under `lattice/herfurtner/`: `01_exponents.py`, `07_classtable.py`, `04_herfurtner.py` (exponents, classes, Herfurtner’s table and its cross-ratios); `02_hscan.c` with `run_scan.sh`, `run_a0.sh` (≈ 75 min, twelve cores), `08_report.py`, `09_table.py` (the scan and its exact re-verification to $n = 300$); `05_jtest.gp`, `10_jtest_all.py`, `11_jall.gp` (the $\mathcal{J}$ -test at $\deg \leq 12$); `13_fastfilter.py`, `14_decay.py` (the complete positive-score pass); `15_k3jtest.gp`, `15_k3rows.py`, `15_k3run.gp` (the raised window, $\text{DMAX} = \text{HMAX} = 24$, $\text{NTERM} = 140$, $\text{GLIM} = 4096$, $\text{KPRIME} = 2^{61} - 1$, an $\mathbb{F}_p$ filter with no false negatives and exact certificates); data in `out/`. Under `lattice/cusp_move/`: `01_general_move.py` (the symbolic proof of Theorem 6.1) through `12_domb_pairs.gp`. Under `lattice/acf_one_surface/`: `01_cusp_move.gp` through `08_census_check.gp`. Under `lattice/noncongruence_scan/`: `01_class_scan.c`, `04_score.py`, `06_groups.py` (the 28 groups, Wohlfahrt test), `02_eta_enum.py`, `03b_exact.gp`, `09_level5.gp`, `10_ident5.gp`, `12_source5.gp`, `14_mirror234.gp`, `16_arctan.py`. Under `lattice/sqrt_aperly/`: `01_sym2.py`, `12_sharp.gp`, `06_fold.gp`. Under `lattice/number_field/`: `nf_herfurtner_uniformisation.gp`, `cfold.gp` with `run_all.gp`, `run_v23.gp`, `run_t.gp`, and the sweeps `hunt3.gp`–`hunt9.gp` (`xi_targets.txt`).

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- [21] Claude (Opus 5), Claude (Fable) and River, *Project notes and scripts* (2026): `HERFURTNER_CLASSIFICATION.md` (the exponent dictionary, the cross-ratio invariant, the scan, the $\mathcal{J}$ -map test), `HERFURTNER_K3_WINDOW.md` (the Riemann–Hurwitz bound and the $\deg \leq 24$ recheck), `CUSP_MOVE_PROGRAM.md` (the cusp move, orbits, rigidity), `ACF_ONE_SURFACE.md` (**A**, **C**, **F** on one surface), `NONCONGRUENCE_SCAN.md` (Theorems N1–N3, the 28 groups, the two scans), `SQRT_APERY.md` (Beukers’ row), `NUMBER_FIELD_PROGRAM.md` (uniformisation of the two new rows, the Chowla–Selberg period), `HOLONOMY_LINDEP.md` (fold-regularity); scripts under `lattice/`, as itemised in §9.